Chapter 8

Preliminary Considerations for Operative Dentistry

"The treatment of a disease may be entirely impersonal…

The care of a patient must be completely personal".

—Francis W Peabody

This chapter addresses routine chair-side preoperative procedures (before actual tooth preparation). These procedures primarily include patient and operator positions as well as isolation of the operating field.

Preoperative Patient and Dental Team Considerations

In preparation for a clinical procedure, it is important to ensure that patient and operator positions are properly selected, instrument exchange between the dentist and the assistant is efficient, proper illumination is present and magnification is used, as needed.

Patient and Operator Positions

Efficient patient and operator positions are beneficial for the welfare of both individuals. A patient who is in a comfortable position is more relaxed, has less muscle tension and is more capable of cooperating with the dentist.

The practice of dentistry is physically demanding and psychologically stressful. Physical problems may arise if appropriate operating positions are neglected.[1] Most restorative dental procedures may be accomplished with the dentist seated. Positions that create unnecessary curvature of the spine or slumping of the shoulders should be avoided. Proper balance and weight distribution on both feet is essential when operating from a standing position. Generally, any uncomfortable or unnatural position that places undue strain on the body should only rarely be used.

Chair and Patient Positions

Chair and patient positions are important considerations. Dental chairs are designed to provide total body support in any chair position. An available chair accessory is an adjustable headrest cushion or an articulating headrest attached to the chair back. A contoured or lounge-type chair provides adequate patient support and comfort. Most chairs also are equipped with programmable operating positions.

The most common patient positions for operative dentistry are almost *supine* or *reclined 45 degrees* (Fig. 8.1). The choice of patient position varies with the operator, the type of procedure and the area of the mouth involved in the operation. In the almost supine position, the patient's head, knees and feet are approximately the same level. The patient's head should not be lower than the feet; the head should be positioned lower than the feet only in an emergency, as when the patient is in syncope.

Operating Positions

Operating positions may be described by the location of the operator or by the location of the operator's arms in relation to patient position.

- A right-handed operator uses essentially three positions—right front, right and right rear. These are sometimes referred to as the *7-o'clock*, *9-o'clock* and *11-o'clock positions* (Fig. 8.2A).
- For a left-handed operator, the three positions are the *left front*, *left* and *left rear positions* or the *5-o'clock*, *3-o'clock* and *1-o'clock* positions.
- A fourth position, *direct rear position* or *12-o'clock position*, has application for certain areas of the mouth. As a rule, the teeth being treated should be at the same level as the operator's elbow. The operating positions described here are for the right-handed operator; the left-handed operator should substitute left for right.

Right Front Position

The right front position facilitates examination and treatment of mandibular anterior teeth (Fig. 8.2B), mandibular



Figure 8.1 Common patient positions. Both positions are recommended for sit-down dentistry. Use depends on the arch being treated. (A) Supine. (B) Reclined 45 degrees.

posterior teeth (especially on the right side) and maxillary anterior teeth. It is often advantageous to have the patient's head rotated slightly towards the operator.

Right Position

In the right position, the operator is directly to the right of the patient (Fig. 8.2C). This position is convenient for operating on the facial surfaces of maxillary and mandibular right posterior teeth and the occlusal surfaces of mandibular right posterior teeth.

Right Rear Position

The right rear position is the *position of choice* for most operations. The operator is behind and slightly to the right of the patient. The left arm is positioned around the patient's head (Fig. 8.2D). When operating from this position, the lingual and incisal (occlusal) surfaces of maxillary teeth are viewed in the mouth mirror. Direct vision may be used on mandibular teeth, particularly on the left side, but the use of a mouth mirror is advocated for visibility, light reflection and retraction.

Direct Rear Position

The direct rear position is used primarily for operating on the lingual surfaces of mandibular anterior teeth. The operator is located directly behind the patient and looks down over the patient's head (Fig. 8.2E).

General Considerations

Several general considerations regarding chair and patient positions are important.

- The operator should not hesitate to *rotate the patient's head* backward or forward or from side to side to accommodate the demands of access and visibility of the operating field. Minor rotation of the patient's head is not uncomfortable to the patient and allows the operator to maintain his or her basic body position.

Clinical Notes

- As a rule, when operating in the maxillary arch, the maxillary occlusal surfaces (i.e. the maxillary occlusal plane) should be oriented approximately perpendicular to the floor.
- When operating in the mandibular arch, the mandibular occlusal surfaces (i.e. the mandibular occlusal plane) should be oriented approximately 45 degrees to the floor.

The operator's face *should not come too close to the patient's face*. The ideal distance is similar to that for reading a book while sitting in an upright position. The vertical position of the patient should be adjusted to allow the operator to maintain optimal ergonomic back and neck posture.

Another important aspect of proper operating position is to *minimize body contact* with the patient. It is not appropriate for an operator to rest forearms on the patient's shoulders or hands on the patient's face or forehead.

The patient's chest should not be used as an instrument tray. From most positions, the left hand should be free to hold the mouth mirror to reflect light onto the operating field, to view the tooth preparation indirectly or to retract the cheek or tongue.

In certain instances, it is more appropriate to retract the cheek with one or two fingers of the left hand than to use a mouth mirror. It is often possible, however, to retract the cheek and reflect light with the mouth mirror at the same time.

When operating for an extended period, the operator may obtain a certain amount of rest and muscle relaxation by changing operating positions. Operating from a single position through the day, especially if standing, produces unnecessary fatigue. Changing positions, if

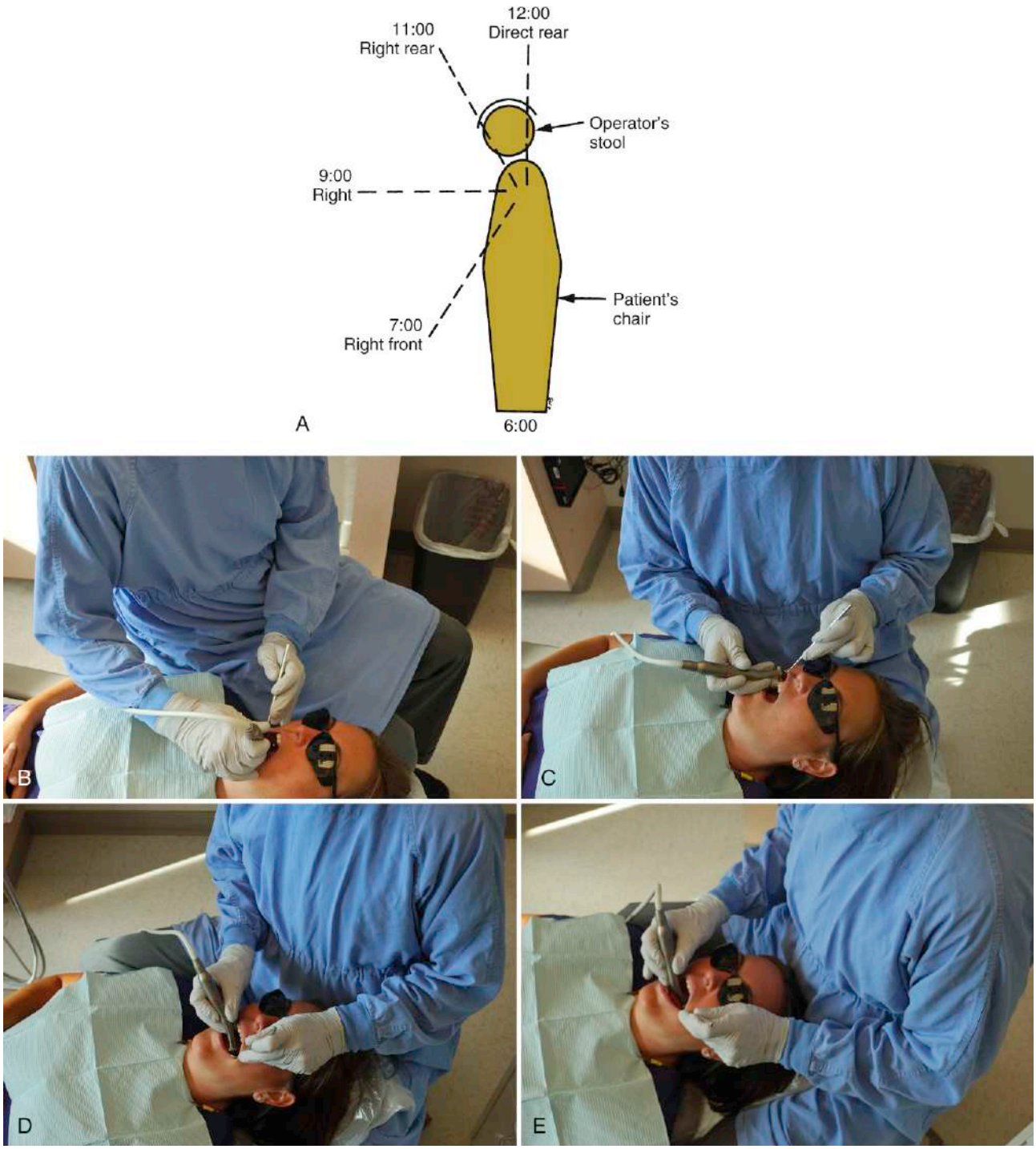


Figure 8.2 Operating positions indicated by arm approach to the patient. (A) Diagrammatic operator positions. (B) Right front. (C) Right. (D) Right rear. (E) Direct rear.
(Courtesy: [B, C, D and E] Dr Mohammad Atieh.)

only for a short time, reduces muscle strain and lessens fatigue.[1]

Operating Stools

A variety of operating stools are available for the dentist and the dental assistant. The seat should be well padded with smooth cushion edges and should be adjustable for optimal leg position and back support. Advantages of the seated work position are compromised if the operator uses the stool improperly.

- The operator should sit back on the cushion, using the entire seat and not just the front edge.
- The upper body should be positioned so that the spinal column is straight or bent slightly forward and supported by the backrest of the stool.



Figure 8.3 Recommended seating positions for operator and chair-side assistant, with the height of the operating field approximately at elbow level of the operator.
(From Robinson DS, Bird DL: *Essentials of dental assisting,* ed 4, St. Louis, MO, 2007, Saunders.)

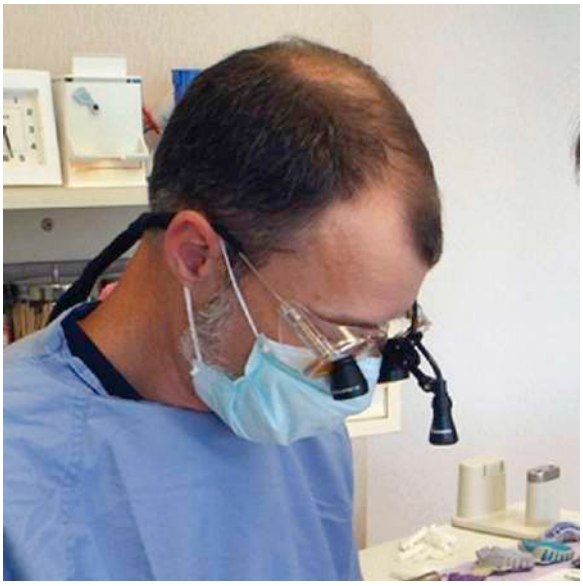


Figure 8.4 Use of magnification with surgical telescopes.

- The thighs should be parallel to the floor, and the lower legs should be perpendicular to the floor. If the seat is too high, its front edge compromises circulation to the user's legs.
- Feet should be flat on the floor.

The seated work position for the assistant is essentially the same as for the operator except that the stool is 4–6 in. higher for maximal visual access. It is important that the assistant's stool have an adequate footrest so that a parallel thigh position is maintained with good foot support. When properly seated, the operator and the assistant are capable of providing dental service throughout the day without an unnecessary decline in efficiency and productivity because of muscle tension and fatigue (Fig. 8.3).

Instrument Exchange

All instrument exchanges between the operator and the assistant should occur in the exchange zone below the patient's chin and a few inches above the patient's chest. Instruments should not be exchanged over the patient's face. During the procedure, the operator should anticipate the next instrument required and inform the assistant accordingly; this allows the instrument to be brought into the exchange zone for a timely exchange.

During proper instrument exchange, the operator should not need to look away from the operating field. The operator should rotate the instrument handle forward to cue the assistant to exchange instruments. The assistant should take the instrument from the operator, rather than the operator dropping it into the assistant's hand, and vice versa. Each person should be sure that the other has a firm grasp on the instrument before it is released.

Magnification and Headlamp Illumination

Another key to the success of clinical operative dentistry is *visual acuity*. The operator must be able to see clearly to attend to the details of each procedure. The use of magnification facilitates attention to detail and does not adversely affect vision. Magnifying lenses have a fixed focal length that often requires the operator to maintain a proper working distance, which helps to ensure good posture. Several types of magnification devices are available, including bifocal eyeglasses, loupes and surgical telescopes (Fig. 8.4). To further improve visual acuity, headlamps are recommended in operative dentistry. Their greatest advantage is the light source being parallel to the clinician's vision, eliminating shadows at the operating field. Current headlamps use light-emitting diode (LED) technology and produce whiter light than conventional tungsten halogen light sources.

Isolation of the Operating Field

The goals of operating field isolation are moisture control, retraction and patient safety.

Goals of Isolation

Moisture Control

Moisture control refers to the exclusion of saliva, gingival sulcular fluid and gingival bleeding from the operating field. It is not possible to properly accomplish operative dentistry without control of mouth moisture. It also involves preventing or limiting the spray from the handpiece

and restorative debris from being swallowed or aspirated by the patient. The rubber dam, suction devices and absorbents are variously effective in controlling moisture.[2,3]

Clinical Notes

This textbook recommends use of the rubber dam as an optimum means of gaining moisture control.

Retraction and Access

The details of a restorative procedure cannot be managed without proper retraction and access. Retraction and access provide maximal exposure of the operating site and usually involve having the patient maintain an open mouth and displacement or retraction of the gingival tissue, tongue, lips and cheek. The rubber dam, high-volume evacuator, absorbents, retraction cord, mouth prop and other isolation devices such as the Isolite (Isolite Systems, Santa Barbara, CA, USA) are used for retraction and access.

Patient Safety

An important consideration of isolating the operating field is the use of means to provide safety (i.e. prevent harm) to the patient during the operation.[4,5] Excessive saliva and handpiece spray may alarm the patient. Small instruments and restorative debris may be aspirated or swallowed. Soft tissue may be iatrogenically damaged. Various isolation techniques and devices limit the potential for adverse outcomes. Harm prevention is achieved as much by the manner in which the devices are used as by the devices themselves.

Local Anaesthesia

Local anesthetics play a role in eliminating the discomfort of dental treatment and controlling moisture by reducing salivary flow. Local anesthetics incorporating a vasoconstrictor also reduce blood flow, which helps control haemorrhage at the operating site.

Rubber Dam Isolation

In 1864 S.C. Barnum, a New York City dentist, introduced the rubber dam into dentistry. Use of the rubber dam ensures appropriate dryness of the teeth and improves the quality of clinical restorative dentistry.[6,7] The rubber dam is used to define the operating field by isolating one or more teeth from the oral environment. The dam eliminates saliva from the operating site and retracts the soft tissue.

Advantages

The advantages of rubber dam isolation of the operating field include

I. *A dry, clean operating field:* For most procedures, rubber dam isolation is the preferred method of obtaining a dry, clean field. The operator is best able to perform procedures such as proper tooth preparation, caries removal and insertion of restorative materials in a dry field. The time saved by operating in a clean field with good visibility may more than compensate for the time spent applying the rubber dam.[8] When excavating a deep caries lesion and risking pulpal exposure, use of the rubber dam is strongly recommended to prevent pulpal contamination from bacteria in oral fluids.

II. *Improved access and visibility:* The rubber dam provides maximal access and visibility. It controls moisture and retracts soft tissue. Gingival tissue is mildly retracted so as to enhance access to and visibility of the gingival aspects of the tooth preparation. The dam also retracts the lips, cheeks and tongue. A dark-coloured rubber dam provides a nonreflective background, which is in contrast to the operating site. The dam allows uninterrupted access and visibility throughout the operative procedure.

III. *Optimization of dental material properties:* The rubber dam prevents moisture contamination and compromise of restorative materials used during the procedure. Amalgam restorative material does not achieve its optimum physical properties if used in a wet field.[6] Bonding to enamel and dentin is severely compromised or nonexistent if the tooth substrate is contaminated with saliva, blood or other oral fluids.[9,10] Some studies have concluded that no difference exists between the use of the rubber dam and cotton roll isolation as long as control of sources of contamination is maintained during the restorative procedures.[2,11–13] However, the effectiveness of rubber dam isolation allows freedom to focus on the details of the restorative procedure which is especially advantageous for those procedures that are technique sensitive.

IV. *Protection of the patient and the operator:* The rubber dam protects the patient and the operator. It protects the patient from aspirating or swallowing small instruments or debris associated with operative procedures.[14] A properly applied rubber dam protects soft tissue and the tongue from irritating or distasteful medicaments (e.g. etching and astringent agents). The dam also offers some soft tissue protection from rotating burs and stones. Authors disagree on whether the rubber dam protects the patient from mercury exposure during amalgam removal.[15,16] However, it is generally agreed that the rubber dam is an effective infection control barrier for the dental office.[17–19]

V. *Operating efficiency:* Use of the rubber dam allows for operating efficiency and increased productivity. Conversation with the patient is limited. The rubber dam retainer (discussed later) helps provide a moderate amount of mouth opening during the procedure. (For additional mouth-opening aids, see Mouth Props.) Quadrant restorative procedures are facilitated. Many

state dental practice acts permit the assistant to place the rubber dam, thus saving time for the dentist. Christensen reported that use of a rubber dam increases the quality and quantity of restorative services.[8]

Disadvantages

Rubber dam use is low among private practitioners.[20–22] Time consumption and patient objection are the most frequently quoted disadvantages of the rubber dam. However, the rubber dam may usually be placed in less than 5 min. The advantages previously mentioned certainly justify any time utilized in accomplishing proper placement.

Certain situations may preclude the use of the rubber dam, including

I. Teeth that have not erupted sufficiently to support a retainer
II. Some third molars
III. Extremely malpositioned teeth

In addition, patients may not tolerate the rubber dam if breathing through the nose is difficult. In rare instances, the patient cannot tolerate a rubber dam because of psychologic reasons or latex allergy.[12,23] Latex-free rubber dam material is, however, currently available (Fig. 8.5). These situations are the exception, and it has been reported that use of the rubber dam was well accepted by most patients and operators.[24]

Materials and Instruments

The materials and instruments necessary for the use of the rubber dam are available from most dental supply companies. It is necessary to have waxed dental tape or floss available so as to lubricate the contact areas of the teeth to be isolated prior to rubber dam placement.



Figure 8.5 Rubber dam material as supplied in sheets.
(From Boyd LRB: *Dental instruments: a pocket guide,* ed 4, St. Louis, MO, 2012, Saunders.)

Rubber Dam Material

Rubber dam material (latex and nonlatex), as with all types of elastic material, will deteriorate over time, resulting in low tear strength.

The dam material is available in 12.5 × 12.5 cm or 15 × 15 cm sheets. The thicknesses or weights available are:

- Thin (0.15 mm)
- Medium (0.2 mm)
- Heavy (0.25 mm)
- Extra heavy (0.30 mm)

Light and dark dam materials are available, and darker colours are generally preferred for contrast. A thicker dam is more effective in retracting tissue and more resistant to tearing; it is especially recommended for isolating Class V lesions in conjunction with a cervical retainer. The thinner material has the advantage of passing through the contacts easier, which is particularly helpful when proximal contacts are broad and tooth mobility is limited.

Clinical Notes

The rubber dam material has a shiny side and a dull side. Because the dull side is less light reflective, it is generally placed facing the occlusal side of the isolated teeth.

Rubber Dam Frame

The rubber dam holder (frame) suspends the borders of the rubber dam. The Young holder is a U-shaped metal frame (Fig. 8.6) with small metal projections for securing the borders of the rubber dam.

Rubber Dam Retainer

The rubber dam retainer consists of four prongs and two jaws connected by a bow (Fig. 8.7). The retainer is used to anchor the dam to the most posterior tooth to be isolated. Retainers also are used to retract gingival tissue. Many

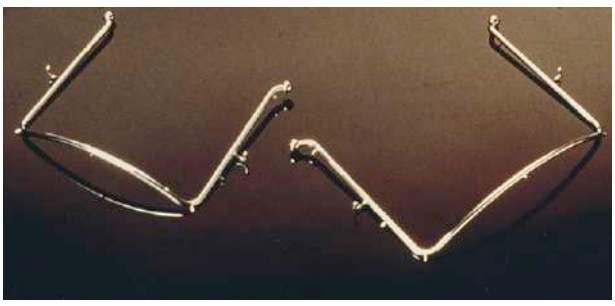


Figure 8.6 Young rubber dam frame (holder).
(From Hargreaves KM, Cohen S: *Cohen's pathways of the pulp,* ed 10, St. Louis, MO, 2011, Mosby.)

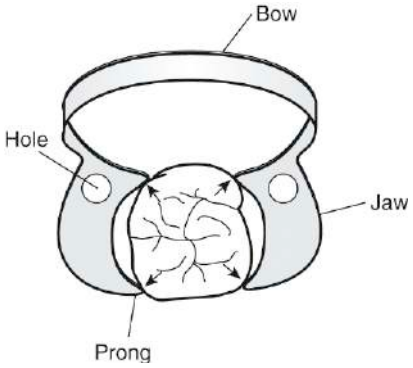


Figure 8.7 Rubber dam retainer. Note four-point prong contact (*arrows*) with tooth.
(From Daniel SJ, Harfst SA, Wilder RS: *Mosby's dental hygiene: concepts, cases, and competencies,* ed 2, St. Louis, MO, 2008, Mosby.)

Table 8.1 Suggested retainers for various anchor tooth applications

Retainer	Application
W56	Most molar anchor teeth
W7	Mandibular molar anchor teeth
W8	Maxillary molar anchor teeth
W4	Most premolar anchor teeth
W2	Small premolar anchor teeth
W27	Terminal mandibular molar anchor teeth requiring preparations involving the distal surface

different sizes and shapes are available, with specific retainers designed for certain teeth (Fig. 8.8). Table 8.1 lists suggested retainer applications. When positioned on a tooth, a properly selected retainer should contact the tooth on its four line angles (Fig. 8.7). This four-point contact prevents rocking or tilting of the retainer. Movement of the retainer on the anchor tooth may injure the gingiva and the tooth, resulting in postoperative soreness or sensitivity.[25] The prongs of some retainers are gingivally directed (inverted) and are helpful when the anchor tooth is only partially erupted or when additional soft tissue retraction is indicated (Fig. 8.9).

Wingless and *winged retainers* are available (Fig. 8.8). The winged retainer has anterior and lateral wings (Fig. 8.10). The wings are designed to provide extra retraction of the rubber dam from the operating field and to allow attachment of the dam to the retainer before conveying the retainer (with dam attached) to the anchor tooth, after which the dam is removed from the lateral wings. As seen in Fig. 8.10, the anterior wings may be removed if they are not desired.

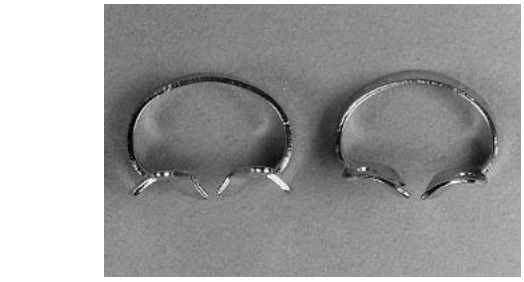


Figure 8.9 Retainers with prongs directed gingivally are helpful when the anchor tooth is only partially erupted.

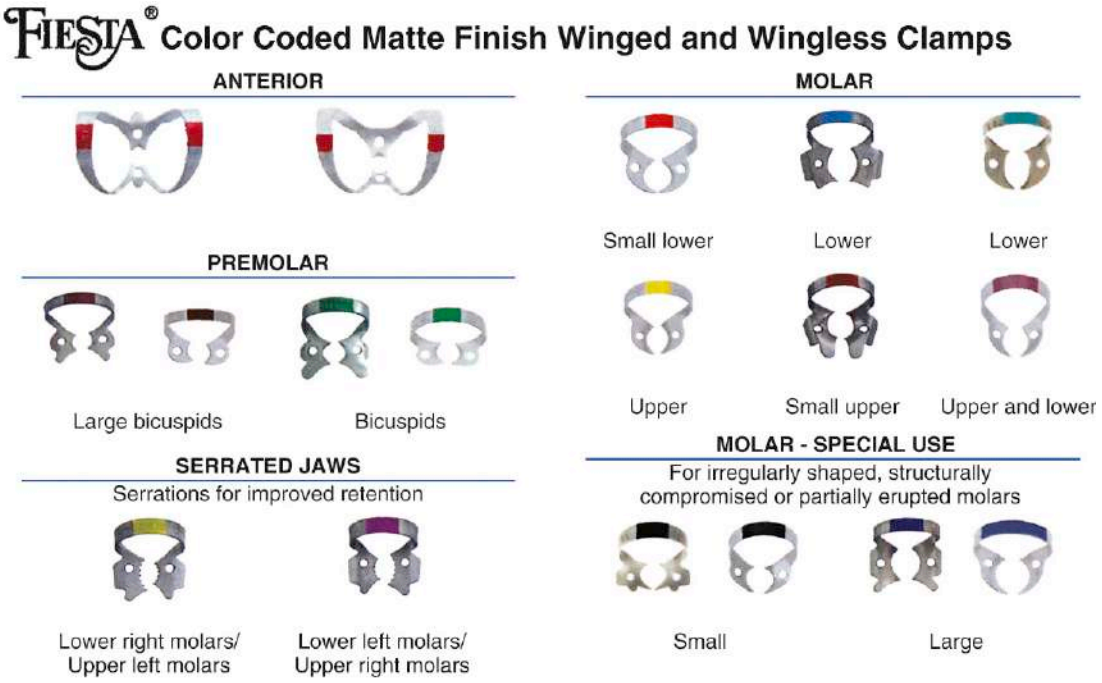


Figure 8.8 Selection of rubber dam retainers. Note retainers with wings. (*Pictured*: Colour-coded matte finish winged and wingless clamps.)
(Courtesy: Coltène/Whaledent Inc., Cuyahoga Falls, OH.)



Figure 8.10 Removing anterior wings (a) on molar retainer. Lateral wings (b) are for attachment of the rubber dam material during placement.

The bow of the retainer (except the No. 212, which is applied after the rubber dam is in place) should be tied with dental floss (Fig. 8.11) approximately 30 cm in length before the retainer is placed in the mouth. For maximal protection, the tie may be threaded through both holes in the jaws of the retainer because the bow of the retainer may fatigue and fracture after multiple uses. The floss allows retrieval of the retainer or its broken parts if they are accidentally swallowed or aspirated. It is sometimes necessary to recontour the jaws of the retainer to the shape of the tooth by grinding with a mounted stone or other cutting instrument (Fig. 8.12). A retainer usually is not required when the dam is applied for treatment of anterior teeth except for the cervical retainer for Class V restorations.

Clinical Notes

The jaws of the retainer should not extend beyond the mesial and distal line angles of the tooth because:

- they may interfere with matrix and wedge placement,
- gingival trauma is more likely to occur,
- a complete seal around the anchor tooth is more difficult to achieve.



Figure 8.12 Recontouring jaws of retainer with mounted stone.

Rubber Dam Punch

The rubber dam punch is a precision instrument having a rotating metal table disc (cutting table) with holes of varying sizes and a tapered, sharp-pointed plunger (Fig. 8.13). Care should be exercised when changing from one hole to another. The plunger should be centred in the cutting hole so that the edges of the holes are not at risk of being



Figure 8.13 Rubber dam punches.
(From Boyd LRB: *Dental instruments: a pocket guide,* ed 4, St. Louis, MO, 2012, Saunders.)

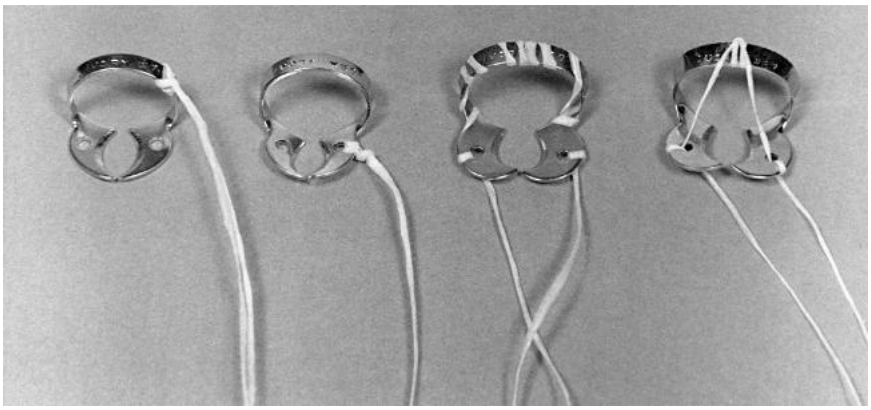


Figure 8.11 Methods of tying retainers with dental floss.

chipped by the plunger tip when the plunger is closed. If the holes in the disk are damaged the cutting quality of the punch is compromised, as evidenced by incompletely cut holes. These holes tear easily when stretched during application over the retainer or tooth.

Retainer Forceps

The rubber dam retainer forceps is used for placement and removal of the retainer from the tooth (Fig. 8.14).

Napkin

The rubber dam napkin, placed between the rubber dam and the patient's skin, has the following benefits (Fig. 8.15):

I. Improvement of patient comfort by reducing direct contact of the rubber material with the skin.
II. Absorption of saliva seeping at the corners of the mouth.
III. Serves as a cushion for the rubber material.
IV. Provides a convenient method of wiping the patient's lips upon removal of the dam.

Lubricant

A water-soluble lubricant applied in the area of the punched holes facilitates the passing of the dam septa through the proximal contact areas of the teeth to be isolated. Rubber dam lubricants are commercially available; however, other lubricants such as shaving cream also are satisfactory.

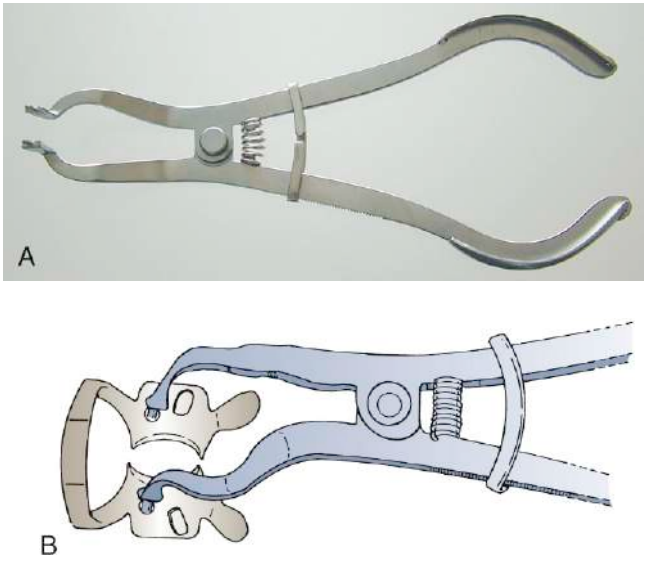


Figure 8.14 Rubber dam forceps (A) engaging retainer (B).
([A] From Boyd LRB: *Dental instruments: a pocket guide,* ed 4, St. Louis, MO, 2012, Saunders. [B] From Baum L, Phillips RW, Lund MR: *Textbook of operative dentistry,* ed 3, Philadelphia, PA, 1995, Saunders.)



Figure 8.15 Disposable rubber dam napkin.
(Courtesy: Coltène/Whaledent Inc., Cuyahoga Falls, OH.)

Additionally, the use of waxed floss enables initial lubrication of contact areas of the teeth to be isolated. Cocoa butter or petroleum jelly may be applied at the corners of the patient's mouth to prevent irritation. These two materials are not satisfactory rubber dam lubricants, however, because both are oil-based and not easily rinsed from the dam and adjacent tooth structure when the dam is placed.

Anchors (Other Than Retainers)

Other anchors, in addition to conventional rubber dam retainers, may also be used. The intensity of the proximal contact itself may be sufficient to anchor the dam on the tooth farthest from the posterior retainer (in the isolated field), eliminating the need for a second retainer (see Step 13 of Procedure 8.1 later in the chapter). To secure the dam further anteriorly or to anchor the dam on any tooth where a retainer is contraindicated, a small piece of rubber dam material (cut from a sheet of dam) or waxed dental tape (or floss) or a rubber Wedjet (Hygenic, Akron, OH, USA) may be passed through the proximal contact. The cut piece of dam material is first stretched, passed through the contact and then released (Fig. 8.16A). When waxed dental tape or floss is used, it should be passed through the contact, looped and passed through a second time (Fig. 8.16B). When the anchor is in place, the dam material, the tape (floss) or Wedjet should be trimmed to prevent interference with the operating site.

Hole Size and Position

Successful isolation of teeth and maintenance of a dry, clean operating field largely depend on hole size and position in the rubber dam.[26] Holes should be punched by following the arch form, making adjustments for malpositioned or missing teeth. Most rubber dam punches have either five or six holes in the cutting table. The smaller

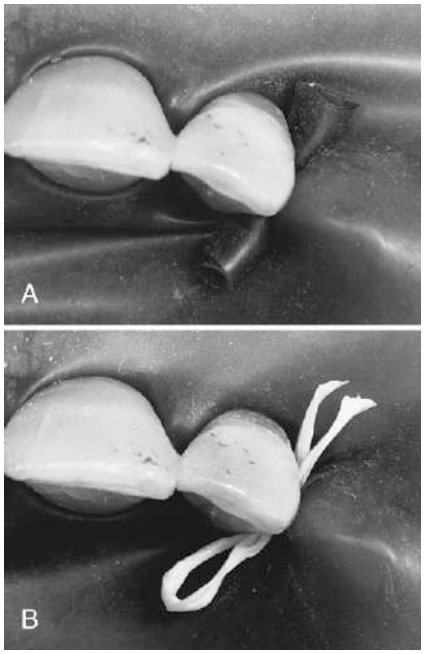


Figure 8.16 (A) Anchor formed from rubber dam material. (B) Anchor formed from dental tape.

holes are used for the incisors, canines and premolars and the larger holes for the molars. The largest hole generally is reserved for the posterior anchor tooth to allow the rubber dam material to stretch over the retainer without tearing (Fig. 8.17). The following guidelines and suggestions (Box 8.1) may be helpful when positioning the holes (Figs. 8.18–8.20).

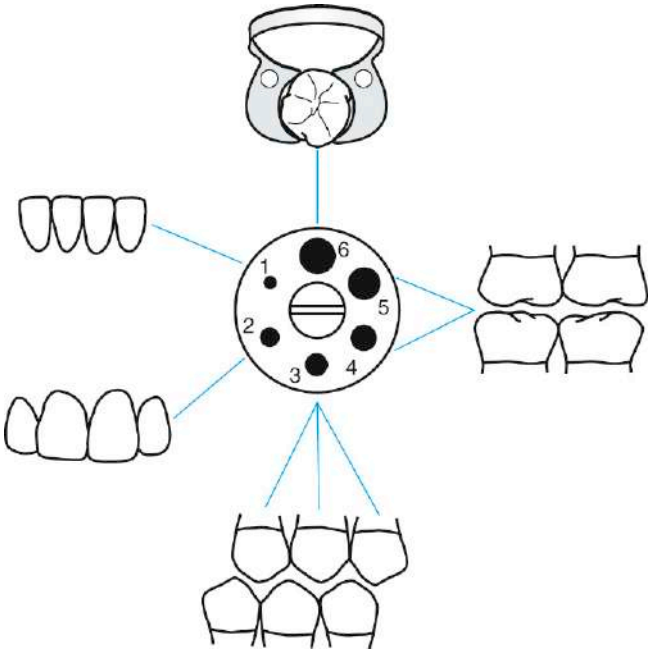


Figure 8.17 Cutting table on rubber dam punch, illustrating use of hole size.
(Modified from Daniel SJ, Harfst SA, Wilder RS: *Mosby's dental hygiene: concepts, cases, and competencies,* ed 2, St. Louis, MO, 2008, Mosby.)

Until these guidelines and suggestions related to hole position are mastered, an inexperienced operator may choose to use commercial products to aid in locating hole position (Fig. 8.21). A rubber stamp that imprints permanent and primary arch forms on the rubber dam is available, and several sheets of dam material may be stamped in advance. A plastic template also may be used to mark hole position. Experienced operators and assistants may not require these aids, and accurate hole location is best achieved by noting the patient's arch form and tooth position.

Placement

Administration of the local anesthetic precedes application of the rubber dam. Peripheral anaesthesia in the area of the procedure allows for more comfortable retainer placement on the anchor tooth. Occasionally, the posterior anchor tooth in the maxillary arch may need to be anesthetized if it is remote from the anaesthetized operating site. The onset of profound anaesthesia will usually occur while the rubber dam is being placed.

The technique for the application of the rubber dam has been presented by numerous authors.[7,27,28] The step-by-step application and removal of the rubber dam, using the maxillary left first molar for the posterior retainer and including the maxillary right lateral incisor as the anterior anchor, is described and illustrated here. The procedure is described as if the operator and the assistant are working together.

Procedure 8.1 demonstrates sequential placement of the retainer and the dam. This approach provides for maximal visibility when placing the retainer, which reduces the risk of impinging on gingival tissue. Isolating a greater number of teeth, as illustrated in Procedure 8.1, is indicated for quadrant operative procedures. For limited operative procedures, it is often acceptable to isolate fewer teeth. Appropriate seal of each tooth is accomplished by inversion of the rubber material in a gingival direction. Interproximal inversion is accomplished first by using dental floss. Inversion of the dam on the facial and lingual surfaces is accomplished by air-drying the surfaces and use of a blunt instrument (Procedure 8.1, Step 18). Procedure 8.2 demonstrates the sequential removal of the dam.

Alternative and Additional Methods and Factors

Applying the Dam and Retainer Simultaneously

The retainer and dam may be placed simultaneously to reduce the risk of the retainer being swallowed or aspirated

Box 8.1 Guidelines for rubber dam-hole positioning

- (Optional) Punch an identification hole in the upper left (i.e. the patient's left) corner of the rubber dam for ease of location of that corner when applying the dam to the holder.
- When operating on the incisors and mesial surfaces of canines, isolate from first premolar to first premolar. Metal retainers usually are not required for this isolation (Fig. 8.18A). If additional access is necessary after isolating teeth, as described, a retainer may be positioned over the dam to engage the adjacent nonisolated tooth, but care must be exercised not to pinch the gingiva beneath the dam (Fig. 8.18B and C).
- When operating on a canine, it is preferable to isolate from the first molar to the opposite lateral incisor. To treat a Class V lesion on a canine, isolate posteriorly to include the first molar to provide access for placement of the cervical retainer on the canine.
- When operating on posterior teeth, isolate anteriorly to include the lateral incisor on the opposite side of the arch from the operating site. In this case, the hole for the lateral incisor is the most remote from the hole for the posterior anchor tooth. Anterior teeth included in the isolation provide finger rests on dry teeth and better access and visibility for the operator and the assistant.
- When operating on premolars, punch holes to include one to two teeth distally and extend anteriorly to include the opposite lateral incisor.
- When operating on molars, punch holes as far distally as possible and extend anteriorly to include the opposite lateral incisor.
- Isolation of a minimum of three teeth is recommended except when endodontic therapy is indicated, and in that case only the tooth to be treated is isolated. The number of teeth and the tooth surfaces to be treated influence the pattern of isolation.
- The distance between holes is equal to the distance from the centre of one tooth to the centre of the adjacent tooth, measured at the level of the gingival tissue. When the distance between holes is excessive, the dam material is excessive and wrinkles between teeth which impede visibility of the proximal surfaces. Conversely, too little distance between holes causes the dam to stretch, resulting in an open space between the rubber material and the isolated tooth and subsequent leakage. When the distance is correct, the dam intimately adapts and isolates the teeth as well as covers and slightly retracts the interdental tissue.
- When the rubber dam is applied to maxillary teeth, the first holes punched (after the optional identification hole) are for the central incisors. These holes are positioned approximately 25 mm from the superior border of the dam (Figs. 8.19A and 8.20), providing sufficient material to cover the patient's upper lip. For a patient with a large upper lip or moustache, position the holes more than 25 mm from the edge. Conversely, for a child or an adult with a small upper lip, the holes should be positioned less than 25 mm from the edge. The holes for the incisors are punched first, followed by the remaining holes as indicated for the anticipated procedure.
- When the rubber dam is applied to mandibular teeth, the first hole punched (after the optional identification hole) is for the posterior anchor tooth that is to receive the retainer. To determine proper location, mentally divide the rubber dam into three vertical sections: left, middle and right. If the anchor tooth is the mandibular first molar, punch the hole for this tooth at a point halfway from the superior edge to the inferior edge and at the junction of the right (or left) and middle thirds (Fig. 8.19B). If the anchor tooth is the second or third molar, the position for the hole moves towards the inferior border and slightly towards the centre of the rubber dam compared with the first molar hole just described (Fig. 8.19C and D). If the anchor tooth is the first premolar, the hole is placed towards the superior border compared with the hole for the first molar and towards the centre of the dam (Fig. 8.19E). The farther posterior the mandibular anchor tooth, the more dam material is required to come from behind the retainer over the upper lip. Fig. 8.20 illustrates the difference in the amount of dam required, comparing the first premolar and the second molar as anchor teeth. The distances also may be compared by noting the length of dam between the superior edge of the dam and the position of the hole for the posterior anchor tooth (Fig. 8.19 B–F).
- When a thinner rubber dam is used, smaller holes must be punched to achieve an adequate seal around the teeth because the thin dam has greater elasticity.

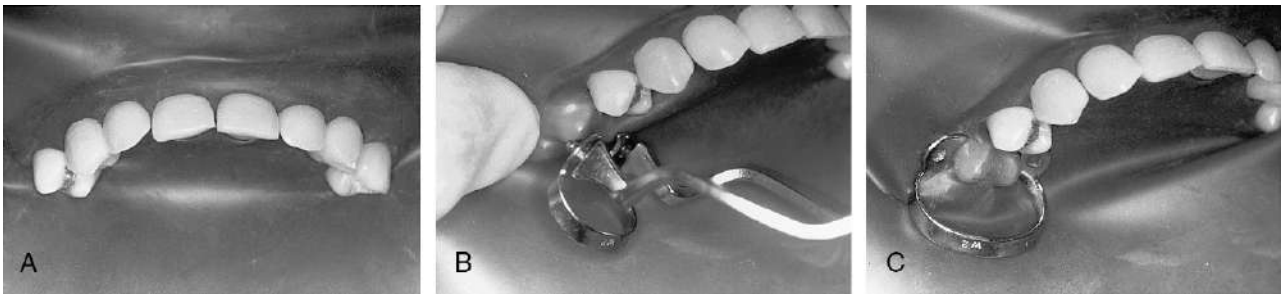


Figure 8.18 (A) Isolation for operating on incisors and mesial surface of canines. (B and C) Increasing access by application of metal retainer over dam and adjacent nonisolated tooth.

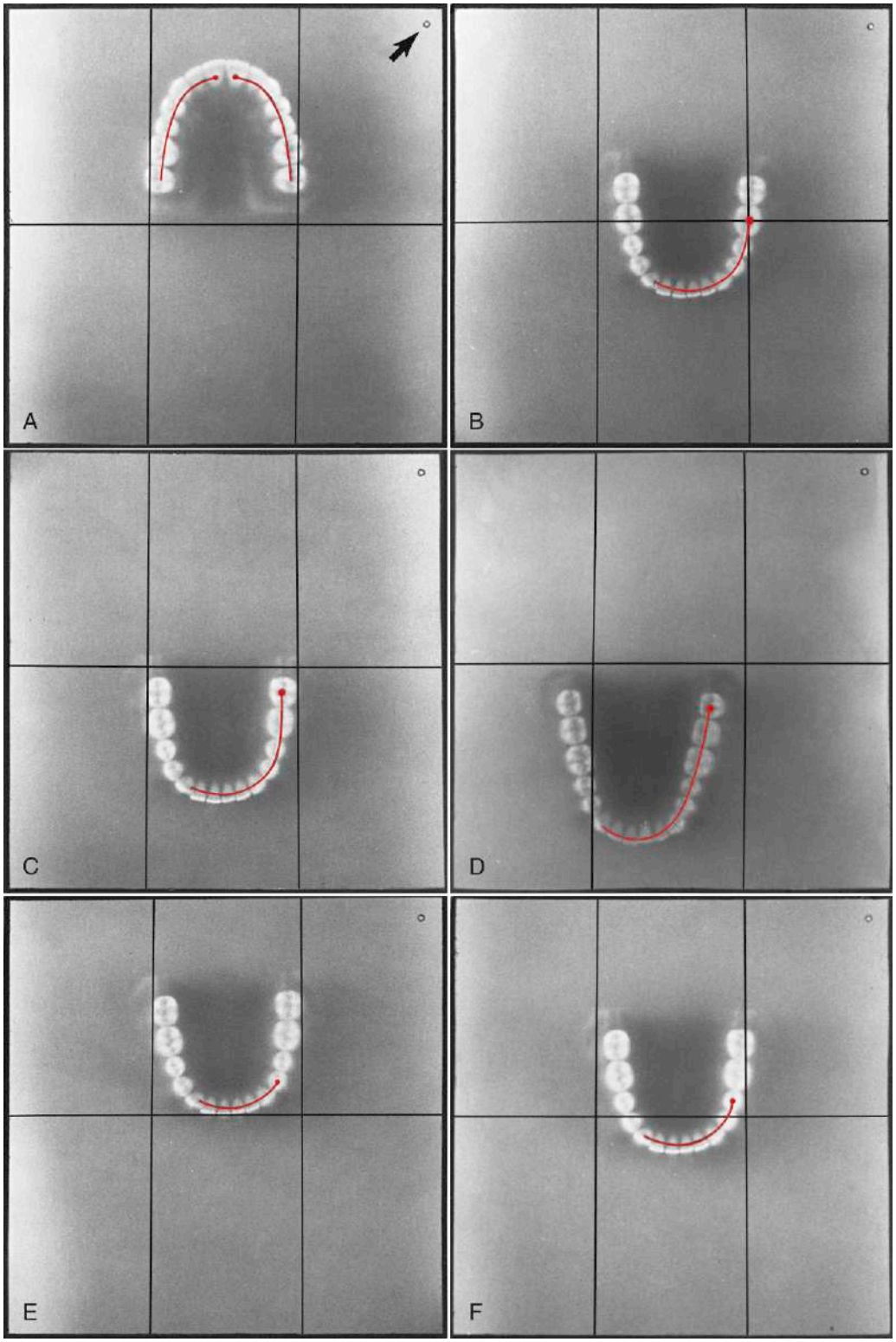


Figure 8.19 Hole position. (A) When maxillary teeth are to be isolated, the first holes punched are for central incisors, approximately 2.5 cm from superior border. (B) Hole position when the anchor tooth is the mandibular first molar. (C) Hole position when the anchor tooth is the mandibular second molar. (D) Hole position when the anchor tooth is the mandibular third molar. (E) Hole position when the anchor tooth is the mandibular first premolar. (F) Hole position when the anchor tooth is the mandibular second premolar. Note the hole punched in each of these six representative rubber dam sheets for identification of the upper left corner (*arrow* in A).

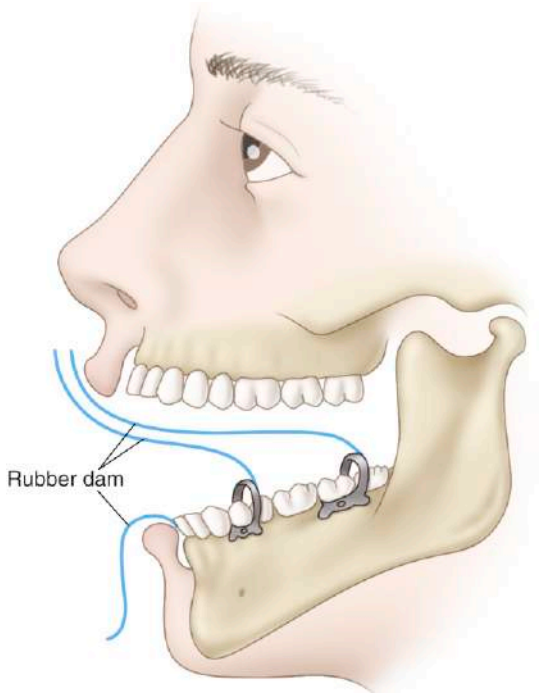


Figure 8.20 The more posterior the mandibular anchor tooth, the more dam material is required to come from behind retainer over the upper lip.

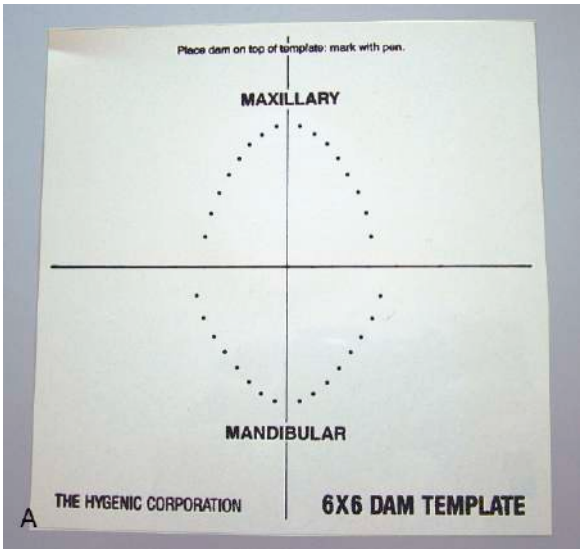




Figure 8.21 Commercial products to aid in locating hole position. (A) Dental dam template. (B) Dental dam stamp.
(From Boyd LRB: *Dental instruments: a pocket guide,* ed 4, St. Louis, MO, 2012, Saunders.)

before the dam is placed (Fig. 8.22). This approach also solves the occasional difficulty of trying to pass the dam over a previously placed retainer, the bow of which is pressing against oral soft tissues. In this method the posterior retainer is applied first to verify a stable fit. The operator removes the retainer and, still holding the retainer with forceps, passes the bow through the proper hole from the underside of the dam (the lubricated rubber dam is held by the assistant) (Fig. 8.22A). The free end of the floss tie should be threaded through the anchor hole before the retainer bow is inserted. When using a retainer with lateral wings, place the retainer in the hole punched for the anchor tooth by stretching the dam to engage these wings (Fig. 8.23).

The operator grasps the handle of the forceps in the right hand and gathers the dam with the left hand to clearly visualize the jaws of the retainer and facilitate its placement (Fig. 8.22B). The operator conveys the retainer (with the dam) into the mouth and positions it on the anchor tooth. Care is needed when applying the retainer to prevent the jaws of the retainer from sliding gingivally and impinging on the soft tissue (Fig. 8.22C).

The assistant gently pulls the inferior border of the dam towards the patient's chin, while the operator positions the superior border over the upper lip. As the assistant holds the borders of the dam, the operator uses the second or middle finger of both hands, one finger facial and the other finger lingual to the bow, to pass the anchor hole borders over and under the jaws of the retainer (Fig. 8.22D). At this point, the application procedure continues as was previously described, beginning with Step 7 in Procedure 8.1.

Applying the Dam Before the Retainer

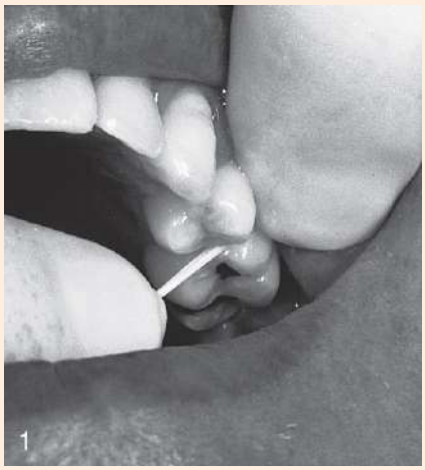
The dam may be stretched over the anchor tooth before the retainer is placed. The advantage of this method is that it is not necessary to manipulate the dam over the retainer. The operator places the retainer, while the dental assistant stretches and holds the dam over the anchor tooth (Fig. 8.24). The disadvantage is the reduced visibility of underlying gingival tissue, which may become impinged on by the retainer.

Procedure 8.1 Application of rubber dam isolation

The application procedure is described for right-handed operators. Left-handed users should change right to left. Each step number has a corresponding illustration.

Step 1: Testing and lubricating the proximal contacts

The operator receives the dental floss from the assistant to test the interproximal contacts and remove debris from the teeth to be isolated. Passing (or attempting to pass) the floss through the contacts identifies any sharp edges of restorations or enamel that must be smoothed or removed to prevent tearing the dam. Using waxed dental floss (or tape) may lubricate tight contacts to facilitate dam placement. Tight contacts that are difficult to floss but do not cut or fray the floss may be wedged apart slightly to permit placement of the rubber dam. A blunt hand instrument may be used for separation. For some clinical situations, the occlusal embrasure above the proximal portion of the tooth to be restored may need to be partially prepared (opened) to eliminate a sharp or difficult contact before the dam is placed.



Step 1: Testing and lubricating the proximal contacts.

Step 2: Punching holes

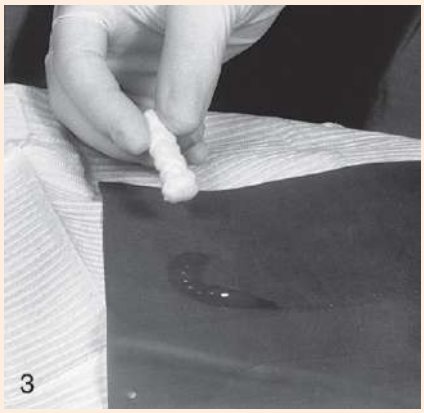
It is recommended that the assistant punch the holes after assessing the arch form and tooth alignment.
Some operators, however, prefer to have the assistant prepunch the dam using holes marked by a template or a rubber dam stamp.



Step 2: Punching the holes.

Step 3: Lubricating the dam

The assistant lubricates both sides of the rubber dam in the area of the punched holes using a cotton roll or gloved fingertip to apply the water-soluble lubricant. This facilitates passing the rubber dam through the contacts. The lips and especially the corners of the mouth may be lubricated with water-insoluble petroleum jelly or cocoa butter to prevent irritation.



Step 3: Lubricating the dam.

Step 4: Selecting the retainer

The operator receives (from the assistant) the rubber dam retainer forceps with the selected retainer and floss tie in position (*A*). The free end of the tie should exit from the cheek side of the retainer. The retainer is placed on the tooth to verify retainer stability. If the retainer fits poorly, it is removed either for adjustment or for selection of a different size.[24] (Retainer adjustment, if needed to provide stability, is discussed in the previous Section on Rubber Dam Retainers.) Whenever the forceps is holding the retainer, care should be taken not to open the retainer more than necessary to secure it in the forceps. Stretching the retainer open for extended periods causes it to lose its elastic recovery. Retainers that have been deformed (sprung) in this way, such as the one shown in *B*, should be discarded.

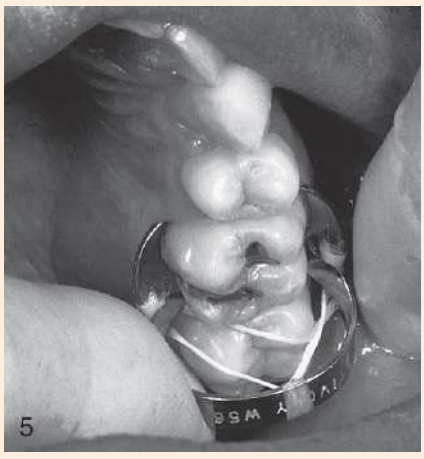
Procedure 8.1 **Application of rubber dam isolation (*cont.*)**



Step 4: Selecting the retainer.
(From Peterson JE, Nation WA, Matsson L: Effect of a rubber dam clamp (retainer) on cementum and junctional epithelium, *Oper Dent 11*:42–45, 1986.)

Step 5: Testing the retainer's stability and retention

If during trial placement the retainer seems acceptable, remove the forceps. Test the retainer's stability and retention by lifting gently in an occlusal direction with a fingertip under the bow of the retainer or by gently tugging on the floss tie. An improperly fitting retainer rocks or is easily dislodged.



Step 5: Testing the retainer's stability and retention.

Step 6: Positioning the dam over the retainer

Before applying the dam, the floss tie may be threaded through the anchor hole or it may be left on the underside of the dam. With the forefingers, stretch the anchor hole of the dam over the retainer (bow first) and then under the retainer jaws. The lip of the hole must pass completely under the retainer jaws. The forefingers may thin out, to a single thickness, the septal dam for the mesial contact of the retainer tooth and attempt to pass it through the contact, lip of the hole first. The septal dam always must pass through its respective contact in single thickness. If it does not pass through readily, it should be passed through with waxed dental floss later in the procedure.



Step 6: Positioning the dam over the retainer.

Step 7: Applying the napkin

The operator now gathers the rubber dam in the left hand, while the assistant inserts the fingers and thumb of the right (or left) hand through the napkin's opening and grasps the bunched dam held by the operator.



Step 7: Applying the napkin.

Step 8: Positioning the napkin

The assistant pulls the bunched dam through the napkin and positions it on the patient's face. The operator helps by positioning the napkin on the patient's right side. The napkin reduces skin contact with the dam.

(*Continued*)

Procedure 8.1 **Application of rubber dam isolation (*cont.*)**



Step 8: Positioning the napkin.

Step 9: Attaching the frame

The operator unfolds the dam. (If an identification hole was punched, it is used to identify the upper left corner.) The assistant aids in unfolding the dam and, while holding the frame in place, attaches the dam to the metal projections on the left side of the frame. The rubber dam material should first be attached to the area of frame that is located on the same arch that the retainer/anchor tooth is located. This limits the likelihood of retainer dislodgement during rubber dam suspension. This is then followed by suspending the rest of the rubber dam on the frame. The frame is positioned on the outside the dam. The curvature of the frame should be concentric with the patient's face. The dam lies between the frame and napkin. Either the operator or the assistant attaches the dam along the inferior border of the frame. Attaching the dam to the frame at this time controls the dam to provide access and visibility. The free ends of the floss tie are secured to the frame.



Step 9: Attaching the frame.

Step 10 (Optional): attaching the neck strap

The assistant attaches the neck strap to the left side of the frame and passes it behind the patient's neck. The operator attaches it to the right side of the frame. Neck strap tension is adjusted to stabilize the frame and hold the frame (and periphery of the dam) gently against the face and away from the operating field. If desired, using soft tissue paper between the neck and strap may prevent contact of the patient's neck against the strap.



Step 10 (Optional): Attaching the neck strap.

Step 11: Passing the dam through the posterior contact

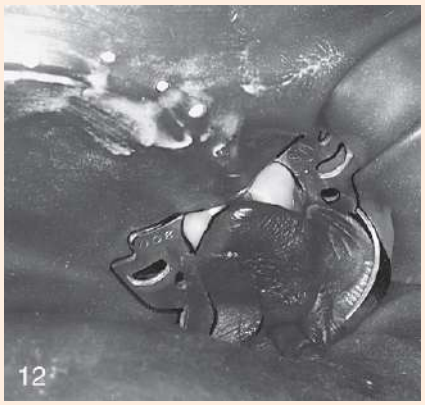
If a tooth is present distal to the retainer, the distal edge of the posterior anchor hole should be passed through the contact (single thickness, with no folds) to ensure a seal around the anchor tooth. If necessary, use waxed dental floss to assist in this procedure (see Step 15 for the use of dental floss). If the retainer comes off unintentionally as this is done or during subsequent procedures, passage of the dam through the distal contact anchors the dam sufficiently to allow easier reapplication of the retainer or placement of an adjusted or different retainer.



Step 11: Passing the dam through the posterior contact.

Step 12 (Optional): Applying a rigid supporting material

If the stability of the retainer is questionable, a rigid supporting material such as a quick-set PVS bite registration material or a low-fusing modelling compound may be applied.



Step 12 (Optional): Applying the rigid material.

Procedure 8.1 **Application of rubber dam isolation (*cont.*)**

Step 13: Applying the anterior anchor (if needed)

The operator passes the dam over the anterior anchor tooth, anchoring the anterior portion of the rubber dam. Usually, the dam passes easily through the mesial and distal contacts of the anchor tooth if it is passed in single thickness starting with the lip of the hole. Stretching the lip of the hole and sliding it back and forth aids in positioning the septum. When the contact farthest from the retainer is minimal (light), an anchor may be required in the form of a double thickness of dental floss or a narrow strip of dam material or Wedjet that is stretched, inserted and released. If the contact is open, a rolled piece of dam material may be used.



Step 13: Applying the anterior anchor (if needed).

Step 14: Passing the septa through the contacts without dental floss

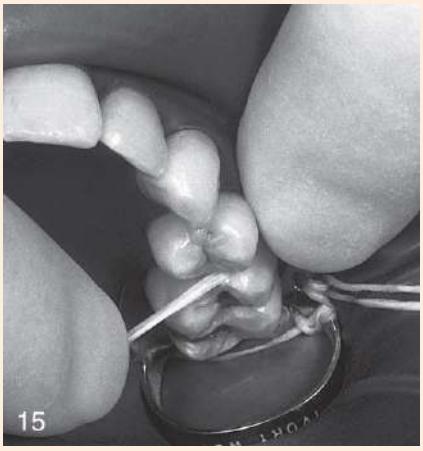
The operator passes the septa through as many contacts as possible without the use of dental floss by stretching the septal dam faciogingivally and linguogingivally with the forefingers. Each septum must not be allowed to bunch or fold. Rather its passage through the contact should be started with a single edge and continued with a single thickness. Passing the dam through as many contacts as possible without using dental floss is urged because the use of dental floss always increases the risk of tearing holes in the septa. Slight separation (wedging) of the teeth is sometimes an aid when the contacts are extremely tight. Pressure from a blunt hand instrument (e.g. beaver-tail burnisher) applied in the facial embrasure gingival to the contact usually is sufficient to obtain enough separation to permit the septum to pass through the contact.



Step 14: Passing the septa through the contacts without dental floss.

Step 15: Passing the septa through the contacts with floss

Use waxed dental floss to pass the dam through the remaining contacts. Dental tape may be preferred over floss because its wider dimension more effectively carries the rubber septa through the contacts and may be less likely to cut the septa. The waxed variety makes passage easier and decreases the chances for cutting holes in the septa or tearing the edges of the holes. The leading edge of the septum should be over the contact, ready to be drawn into and through the contact with dental floss. As before, the septal rubber should be kept in single thickness with no folds. Dental floss should be placed at the contact on a slight angle. With a good finger rest on the tooth, dental floss should be controlled so that it slides (not snaps) through the proximal contact, preventing damage to the interdental tissues. When the leading edge of the septum has passed the contact, the remaining interseptal dam can be carried through more easily.



Step 15: Passing the septa through the contacts with waxed dental floss (or tape).

Step 16 (Optional): Technique for using dental floss

Often, several passes with dental floss are required to carry a reluctant septum through a tight contact. When this happens, previously passed floss should be left in the gingival embrasure until the entire septum has been placed successfully with subsequent passage of dental floss. This prevents a partially passed septum from being removed or torn. The double strand of the floss is removed from the facial embrasure.



Step 16 (Optional): Technique for using dental floss.

(*Continued*)

Procedure 8.1 **Application of rubber dam isolation (*cont.*)**

Step 17: Inverting the dam interproximally

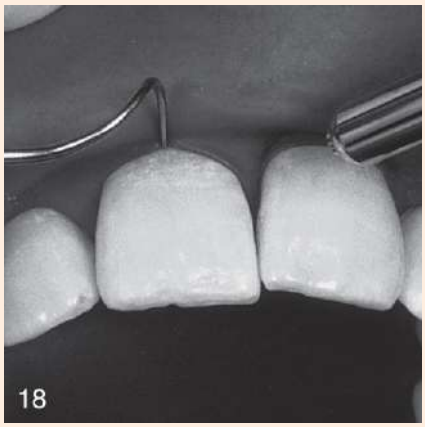
Invert the dam into the gingival sulcus to complete the seal around the tooth and prevent leakage. Often the dam inverts itself as the septa are passed through the contacts as a result of the dam being stretched gingivally. The operator should verify that the dam is inverted interproximally. Inversion in this region is best accomplished with dental tape (or floss).



Step 17: Inverting the dam interproximally.

Step 18: Inverting the dam faciolingually

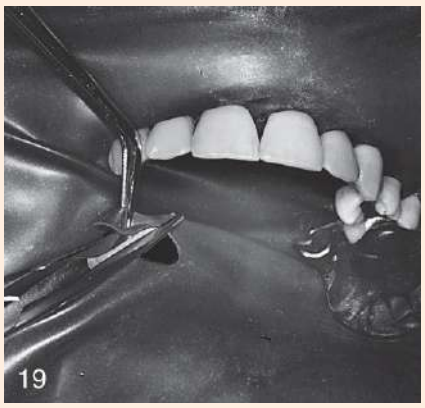
With the edges of the dam inverted interproximally, complete the inversion facially and lingually using an explorer or a beaver-tail burnisher while the assistant directs a stream of air onto the tooth. Move the explorer around the neck of the tooth facially and lingually with the tip perpendicular to the tooth surface or directed slightly gingivally. A dry surface prevents the dam from sliding out of the crevice. Alternatively, the dam may be inverted facially and lingually by drying the tooth while stretching the dam gingivally and releasing it slowly.



Step 18: Inverting the dam faciolingually.

Step 19 (Optional): Using a saliva ejector

The use of a saliva ejector is optional because most patients are able and usually prefer to swallow excess saliva. Salivation is greatly reduced when profound anaesthesia is obtained. If salivation is a problem, the operator or assistant uses cotton pliers to pick up the dam lingual to mandibular incisors and cuts a small hole through which the saliva ejector is inserted. The hole should be positioned so that the rubber dam helps support the weight of the ejector, preventing pressure on the delicate tissues in the floor of the mouth.



Step 19 (optional): Creating a hole for the use of a saliva ejector.

Step 20: Confirming proper application of the rubber dam

The properly applied rubber dam is securely positioned and comfortable to the patient. The patient should be assured that the rubber dam does not prevent swallowing or mouth closing (about halfway) during a pause in the procedure.



Step 20: Confirming proper application of the rubber dam.

Step 21: Checking for access and visibility

Check to see that the completed rubber dam provides maximal access and visibility for the operative procedure.



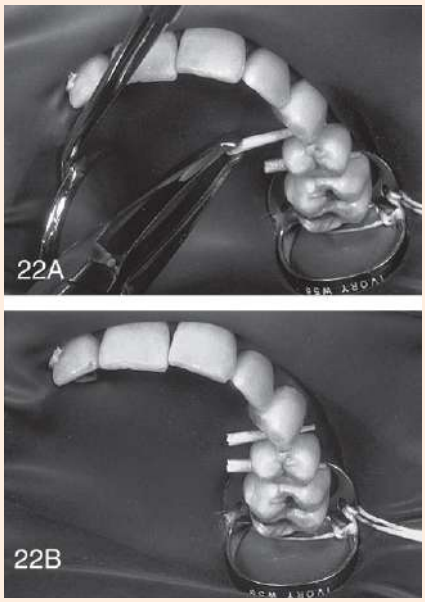
Step 21: Checking for access and visibility.

Procedure 8.1 Application of rubber dam isolation (*cont.*)

Step 22: Inserting the wedges

For proximal surface preparations (Classes II, III and IV), many operators consider the insertion of interproximal wedges as the final step in rubber dam application. Wedges are generally round toothpick ends about 12 mm in length that are snugly inserted into the gingival embrasures from the facial or lingual embrasure, whichever is greater, using No. 110 pliers.

To facilitate wedge insertion, first stretch the dam slightly by fingertip pressure in the direction opposite wedge insertion (*A*), then insert the wedge while slowly releasing the dam. This results in a passive dam under the wedge (i.e. the elastic dam does cause the wedge to rebound) and prevents bunching or tearing of the septal dam during wedge insertion. Lubricating the wedge with water may facilitate wedge placement without rebound. The inserted wedges appear in *B*.



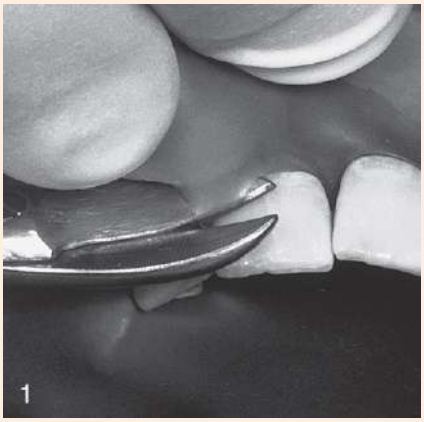
Step 22: Inserting the wedges.

Procedure 8.2 Removal of rubber dam isolation

Before the removal of the rubber dam, rinse and suction away any debris that may have collected to prevent it from falling onto the floor of the mouth during the removal procedure. If a saliva ejector was used, remove it at this time. Each numbered step has a corresponding illustration.

Step 1: Cutting the septa

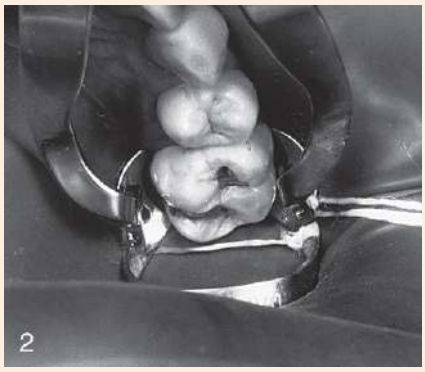
Stretch the dam facially, pulling the septal rubber away from gingival tissue and the tooth. Protect the underlying soft tissue by placing a fingertip beneath the septum. Clip each septum with blunt-tipped scissors, freeing the dam from the interproximal spaces, but leave the dam over the anterior and posterior anchor teeth. To prevent inadvertent soft tissue damage, curved nose scissors are preferred.



Step 1: Cutting the septa.

Step 2: Removing the retainer

Engage the retainer with retainer forceps. It is unnecessary to remove any rigid retaining material, if used, because it will break free as the retainer is spread and lifted from the tooth. While the operator removes the retainer, the assistant releases the neck strap, if used, from the left side of the frame.



Step 2: Removing the retainer.

Step 3: Removing the dam

After the retainer is removed, release the dam from the anterior anchor tooth and remove the dam and frame simultaneously. While doing this, caution the patient not to bite on newly inserted restoration(s) (especially newly placed amalgam) until the occlusion can be evaluated.

(*Continued*)

Procedure 8.2 **Removal of rubber dam isolation (*cont.*)**



Step 3: Removing the dam.

Step 4: Wiping the lips

Wipe the patient's lips with the napkin immediately after the dam and frame are removed. This helps prevent saliva from getting on the patient's face and is comforting to the patient.



Step 4: Wiping the lips.

Step 5: Rinsing the mouth and massaging the tissue

Rinse teeth and the mouth using the air–water spray and the high-volume evacuator. To enhance circulation, particularly around anchor teeth, massage the tissue around the teeth that were isolated.



Step 5: Rinsing the mouth and massaging the tissue.

Step 6: Examining the dam

Lay the sheet of rubber dam over a light-coloured flat surface or hold it up to the operating light to determine that no portion of the rubber dam has remained between or around the teeth. Such a remnant would cause gingival inflammation. Use floss to remove any rubber dam material that remains lodged between the teeth.



Step 6: Examining the dam.

Cervical Retainer Placement

The use of a No. 212 cervical retainer for restoration of Class V tooth preparations was recommended by Markley.[29] When punching holes in the rubber dam, the hole for the tooth to receive this retainer for a facial cervical restoration should be positioned slightly facial to the arch form to compensate for the extension of the dam to the cervical area (Fig. 8.25A). The farther gingivally the lesion extends, the farther the hole must be positioned from the arch form. In addition, the hole should be slightly larger and the distance between it and the adjacent holes should be slightly increased (Fig. 8.26). If the cervical retainer is to be placed on an incisor, isolation should be extended to include the first premolars, and metal retainers usually are not needed to anchor the dam (Fig. 8.25B). If the cervical retainer is to be placed on a canine or a posterior tooth, the anchor tooth retainer is positioned sufficiently posterior so as to not interfere with placement of the cervical retainer. If this is not possible, the anchor tooth retainer should be removed before positioning the cervical retainer. A heavier rubber dam usually is recommended for better tissue retraction for such procedures.

The operator engages the jaws of the cervical retainer with the forceps, spreads the retainer sufficiently and positions its lingual jaw against the tooth at the height of

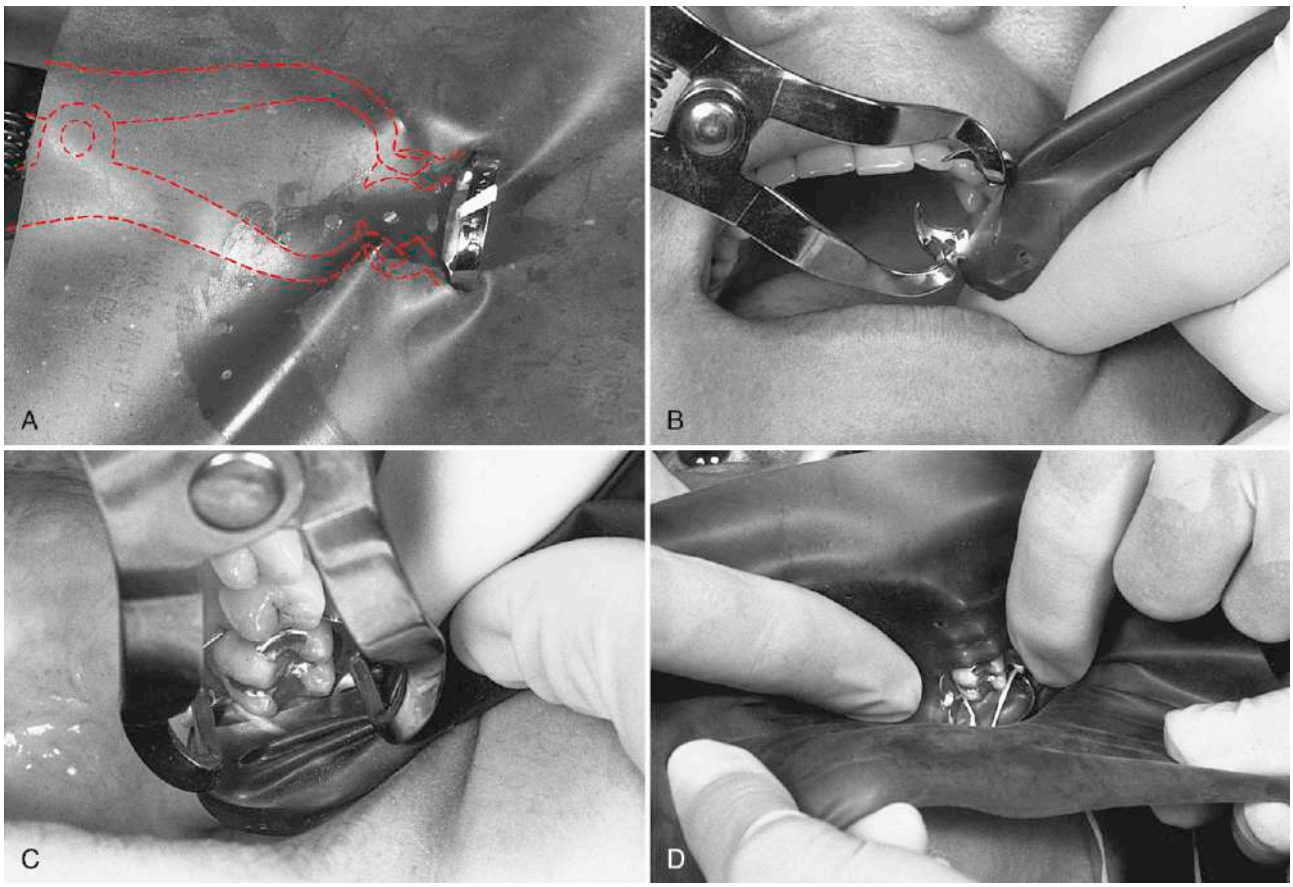


Figure 8.22 (A) Bow being passed through the posterior anchor hole from the underside of the dam. (B) Gathering the dam to facilitate placement of the retainer. (C) Positioning the retainer on the anchor tooth. (D) Stretching the anchor hole borders over and under the jaws of the retainer.

contour (Fig. 8.25C). The operator gently moves the retainer jaw gingivally, depressing the dam and soft tissue, until the jaw of the retainer is positioned slightly apical of the height of contour (Fig. 8.25D). Care should be exercised in not allowing the lingual jaw to pinch the lingual gingiva or injure the gingival attachment. While positioning the lingual jaw, the index finger of the left hand should help in supporting and guiding the retainer jaw gingivally to the proper location.

While stabilizing the lingual jaw with the index finger, the operator uses the thumb of the left hand to pull the dam apically to expose the facial lesion and gingival crest (Fig. 8.25E). The operator positions the facial jaw gingival to the lesion and releases the dam held by the thumb. Next, the operator moves the thumb onto the facial jaw to secure it (Fig. 8.25F). Care should be exercised while positioning the facial jaw so as to not scar enamel or



Figure 8.23 The lip of hole for the anchor tooth is stretched to engage the lateral wings of the retainer.



Figure 8.24 The retainer is applied after the dam is stretched over the posterior anchor tooth.

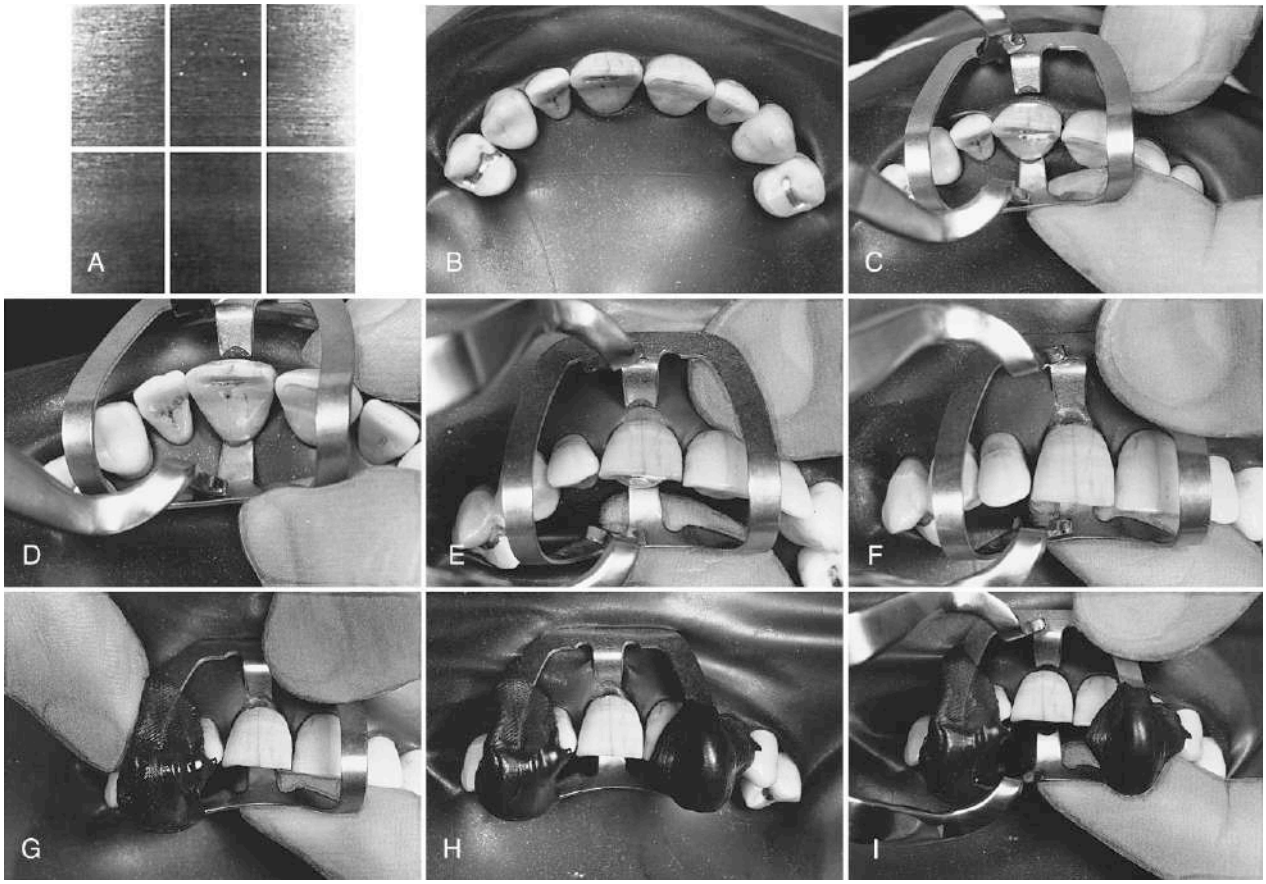


Figure 8.25 Applying a cervical retainer. (A) The hole for maxillary right central incisor is punched facial to the arch form. (B) Isolation is extended to include the first premolars; metal posterior retainers are unnecessary. (C) First position the lingual jaw touching the height of contour, while keeping the facial jaw from touching the tooth; steady the retainer with the fingers of the left hand using the index finger under the lingual bow and the thumb under the facial bow. (D) Note the final position of the lingual jaw after gently moving it apical of height of contour, with fingers continually supporting and guiding the retainer and with the facial jaw away from the tooth. (E) Stretch the facial rubber apically by the thumb to expose the lesion and soft tissue, with the forefinger maintaining the position of the lingual jaw and with the facial jaw not touching. (F) Note the facial jaw having apically retracted the tissue and the dam and in position against the tooth 0.5–1 mm apical of lesion. The thumb has now moved from under the facial bow to apply holding pressure, while the index finger continues to maintain the lingual jaw position. (G) Apply stabilizing material over and under the bow and into the gingival embrasures, while the fingers of left hand hold the retainer's position. (H) Application of the retainer is completed by the addition of a stabilizing material to the other bow and into the gingival embrasures. The retainer holes are accessible to the forceps for removal. (I) Note the removal of the retainer by ample spreading of the retainer jaws before lifting the retainer from the site of the operation.

cementum. The tip of each retainer jaw should not be sharp and should conform to the contour of the engaged tooth surface. The retainer jaw should not be positioned too close to the lesion because of the danger of collapsing carious or weak tooth structure. Such proximity also would limit access and visibility to the operating site. As a rule, the facial jaw should be at least 0.5 mm gingival to the anticipated location of the gingival margin of the completed tooth preparation. While maintaining the retainer's position with the fingers of the left hand, the operator removes the forceps.

At times, the No. 212 retainer needs to be stabilized on the tooth with a fast-setting rigid material (e.g. polyvinyl siloxane [PVS] bite registration material or stick compound) (Fig. 8.25G and H). To remove the cervical retainer, the operator engages it with the forceps, spreads the retainer jaws to free the compound support and lifts the retainer incisally (occlusally), being careful to spread the retainer sufficiently to prevent its jaws from scraping the tooth or damaging the newly inserted restoration (Fig. 8.25I). The embrasures are freed of any remaining PVS or stick compound before removing the rubber dam.

A modified No. 212 retainer is recommended, especially for treatment of cervical lesions with greatly extended gingival margins. The modified No. 212 retainer may be ordered, if specified, or the operator may manually modify an existing No. 212 retainer. The modification technique involves heating each jaw of the retainer in an open flame, then bending it with No. 110 pliers from its oblique orientation to a more horizontal one. Allowing the modified retainer to bench-cool returns it to its original hardened state.

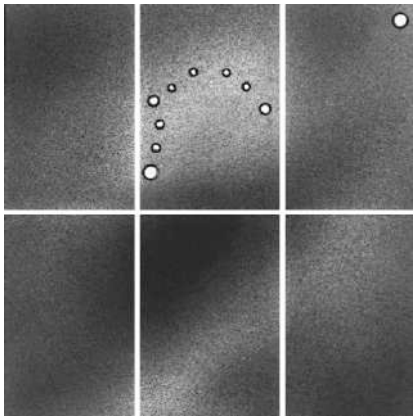


Figure 8.26 The hole position for the tooth (maxillary right canine) to receive the cervical retainer is positioned facially to the arch form.

Fixed Bridge Isolation

It is sometimes necessary to isolate one or more abutment teeth of a fixed bridge. Indications for fixed bridge isolation include restoration of an adjacent proximal surface and cervical restoration of an abutment tooth.

The technique suggested for this procedure[30] is as follows: The rubber dam is punched as usual, except for providing one large hole for each unit in the bridge. Fixed bridge isolation is accomplished after the remainder of the dam is applied (Fig. 8.27A). A blunted, curved suture needle with dental floss attached is threaded from the facial aspect through the hole for the anterior abutment and then under the anterior connector and back through the same hole on the lingual side (Fig. 8.27B). The needle's direction is reversed as it is passed from the lingual side through the hole for the second bridge unit, then under the same anterior connector and through the hole of the second bridge unit on the facial side (Fig. 8.27C). A square knot is tied with the two ends of the floss, pulling the dam material snugly around the connector and into the gingival embrasure. The free ends of the floss should be cut closely so that they neither interfere with access and visibility nor become entangled in a rotating instrument. Each terminal abutment of the bridge is isolated by this method (Fig. 8.27D). If the floss knot on the facial aspect interferes with cervical restoration of an abutment tooth, the operator may tie the septum from the lingual aspect. Removal of the rubber dam isolating a fixed bridge is accomplished by cutting the interseptal rubber over the connectors with scissors and removing the floss ties (Fig. 8.27E). As always, after dam removal, the operator needs to verify that no dam segments are missing and massage the adjacent gingival tissue (as in Procedure 8.2, Step 5).

Substitution of a Retainer with a Matrix

When a matrix band must be applied to the posterior anchor tooth, the jaws of the retainer often prevent proper positioning and wedging of the matrix (Fig. 8.28A). Successful application of the matrix may be accomplished by substituting the retainer with the matrix. Fig. 15.28B–D illustrates this exchange on a mandibular right molar, as the index finger of the operator depresses the rubber dam adjacent to the facial jaw, gingivally and distally and while the

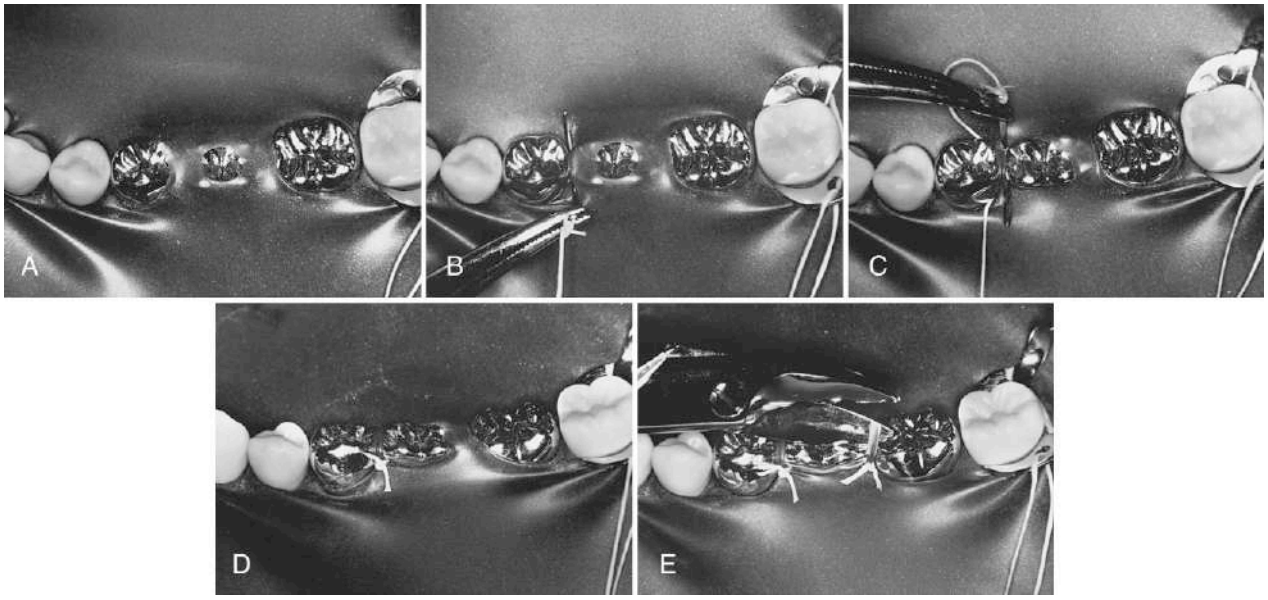


Figure 8.27 Procedure for isolating a fixed bridge. (A) Apply the dam except in the area of the fixed bridge. (B) Thread the blunted suture needle from the facial to the lingual aspect through the anterior abutment hole, then under the anterior connector and back through the same hole on the lingual surface. (C) Pass the needle facially through the hole for the second bridge unit, then under the same connector and through the hole for the second unit. (D) Tie off the first septum. (E) Cut the posterior septum to initiate removal of the dam.

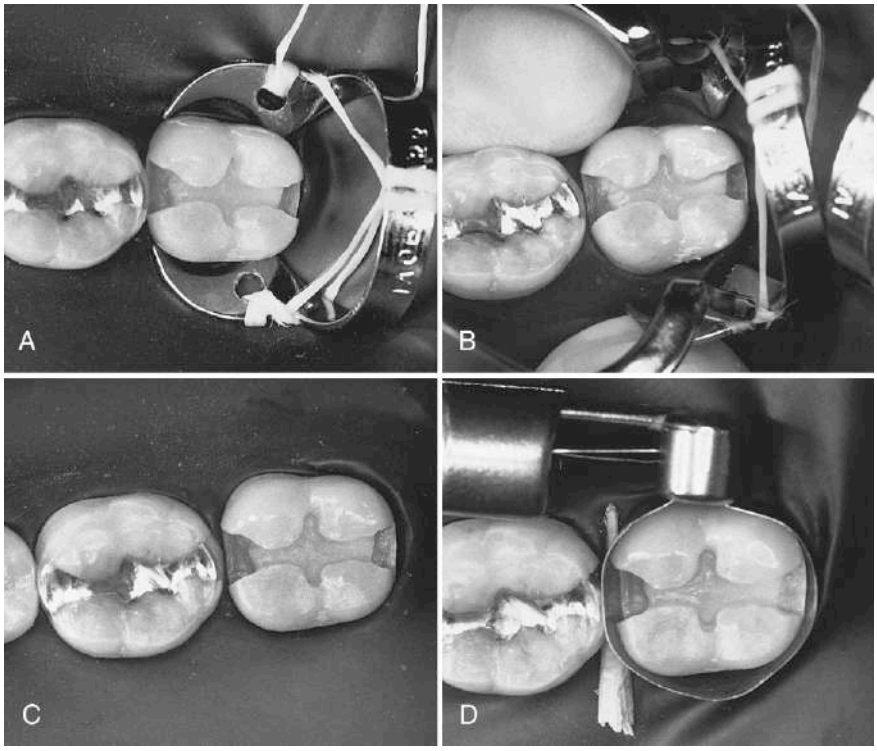


Figure 8.28 Substituting the retainer with matrix on the terminal tooth. (A) Completed preparation of the terminal tooth with the retainer in place. (B) The dentist and the assistant stretch the dam distally and gingivally as the retainer is being removed. (C) The retainer is removed before placement of the matrix. (D) Completed matrix is in place. To maximize access and visibility during insertion, the mouth mirror is used to reflect the dam distally and occlusally.

assistant similarly depresses the dam on the lingual side. After the matrix band is placed, the tension is released on the dam allowing it to invert around the band. The matrix, in contrast to the retainer, has neither jaws nor a bow, so the dam tends to slip occlusally and over the matrix unless dryness is maintained.

The operator obtains access and visibility for insertion of the restorative material by reflecting the dam distally and occlusally with the mirror. Care must be exercised, however, not to stretch the dam so much that it is pulled away from the matrix, permitting leakage around the tooth or slippage over the matrix. After insertion, the occlusal portion is contoured before removing the matrix. To complete the procedure the operator has the choice of removing the matrix, replacing the retainer and completing the contouring or removing the matrix and rubber dam and then completing the contouring of the restoration while using an alternative means of isolation.

Variations with Patient Age

The age of a patient often dictates changes in the procedures of rubber dam application. A few variations are described here. Because young patients have smaller dental arches compared with adult patients, holes should be punched in the dam accordingly. For primary teeth, isolation is usually from the most posterior tooth to the canine on the same side. The sheet of rubber dam may need to be smaller for young patients so that the rubber material does not cover the nose. The unpunched rubber dam is attached to the frame, the holes are punched, the dam with the frame is applied over the anchor tooth and the retainer is applied (Fig. 8.29). Because the dam is generally in place



Figure 8.29 In paediatric dentistry the rubber dam often is attached to a frame before holes are punched. The dam is positioned over the anchor tooth before a retainer is applied (as in Fig. 15.24).

for shorter intervals than in an adult patient, the napkin might not be used.

The jaws of the retainers used on primary and young permanent teeth need to be directed more gingivally because of short clinical crowns or because the anchor tooth's height of contour is below the crest of the gingival tissue.

Isolated teeth with short clinical crowns (other than the anchor tooth) may require ligation with dental floss to hold the dam in position. Generally, ligation is unnecessary if enough teeth are isolated by the rubber dam. When ligatures are indicated, however, a surgeon's knot is used to secure the ligature (Fig. 8.30). The knot is tightened as the ligature is moved gingivally and then secured. Ligatures may be removed by teasing them occlusally with an explorer or by cutting them with a hand instrument or scissors. Ligatures should be removed first during rubber dam removal.

Clinical Notes

- The S.S. White No. 27 retainer is recommended for primary teeth.
- The Ivory No. W14 retainer is recommended for young permanent teeth.

Errors in Application and Removal

Certain errors in application and removal can prevent adequate moisture control, reduce access and visibility or cause injury to the patient.

Off-centre Arch Form

A rubber dam punched off centre (off-centre arch form) may not shield the patient's oral cavity adequately, allowing foreign matter to escape down the patient's throat. An off-centre dam may result in an excess of dam material superiorly that may occlude the patient's nasal airway (Fig. 8.31A). If this happens, the superior border of the dam should be folded under or cut from around the patient's nose (Fig. 8.31B and C). It is important to verify that the rubber dam frame has been applied properly so that the ends of the frame are not dangerously close to the patient's eyes.

Inappropriate Distance Between the Holes

Too little distance between holes precludes adequate isolation because the hole margins in the rubber dam are

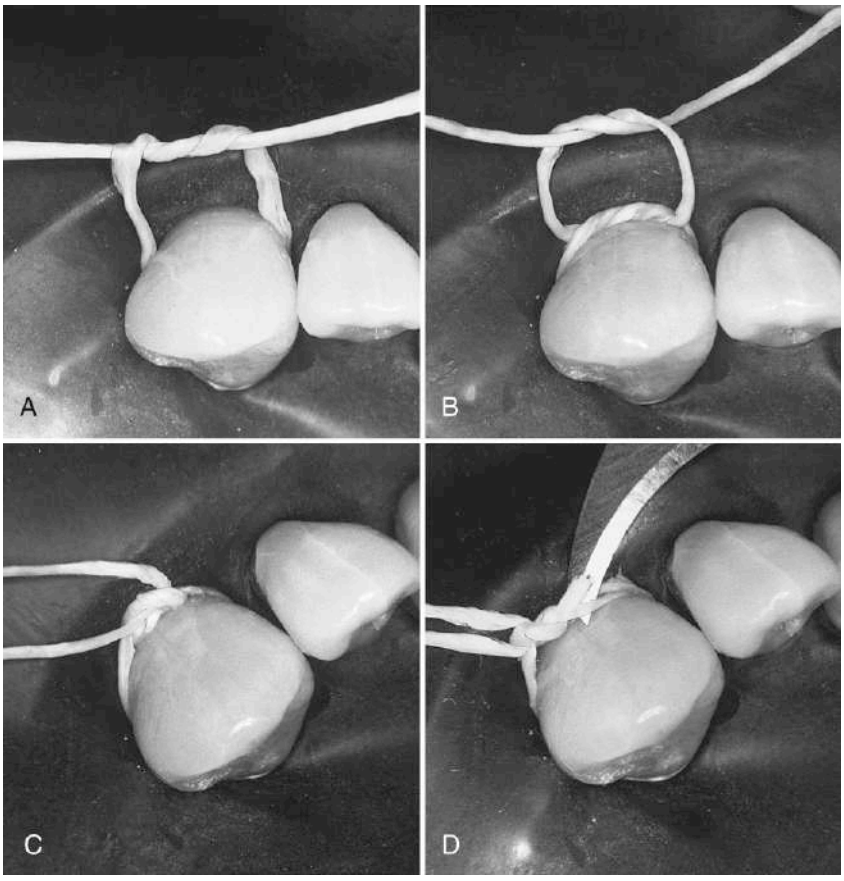


Figure 8.30 Surgeon's knot. (A and B) Dental floss is placed around the tooth gingival to the height of contour (A), and a knot is tied by first making two loops with the free ends, followed by a single loop (B). (C) The free ends are not cut but tied to frame to serve as a reminder that ligature is in place. (D) To remove the ligature, simply cut the tape with a scalpel blade, amalgam knife or scissors.

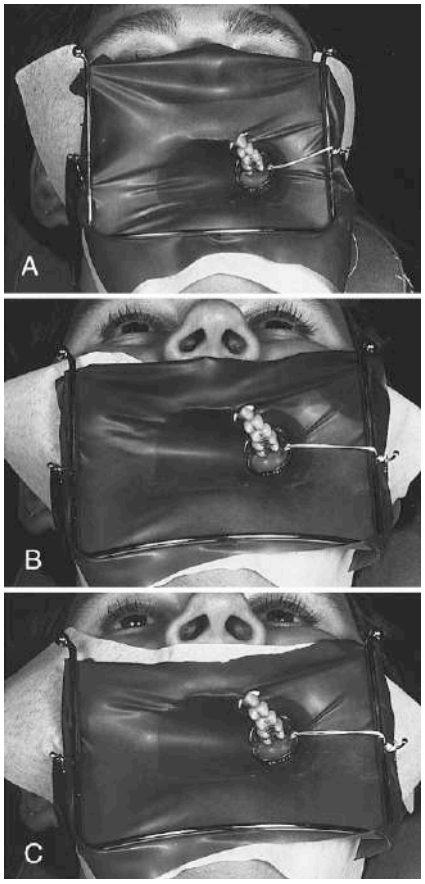


Figure 8.31 (A) An inappropriately punched dam may occlude the patient's nasal airway. (B) Excess dam material along the superior border is folded under to the proper position. (C) Excess dam material is cut from around the patient's nose.

stretched and do not fit snugly around the necks of the teeth. Conversely, too much distance results in excess septal width, causing the dam to wrinkle between the teeth, interfere with proximal access and provide inadequate tissue retraction.

Incorrect Arch Form of Holes

If the punched arch form is too small (incorrect arch form), the holes are stretched open around the teeth, permitting leakage. If the punched arch form is too large, the dam wrinkles around the teeth and may interfere with access.

Inappropriate Retainer

An inappropriate retainer may

I. Be too small, resulting in deformation or breakage when the retainer jaws are overspread;
II. Be unstable on the anchor tooth;
III. Impinge on soft tissue or
IV. Impede wedge placement.

An appropriate retainer should maintain a stable four-point contact with the anchor tooth and not interfere with wedge placement.

Tissue Trauma from Retainer

The jaws and prongs of the rubber dam retainer usually slightly depress, but should not traumatize (puncture, lacerate) the gingiva.

Shredded or Torn Dam

Care should be exercised to prevent shredding or tearing the dam, especially during hole punching or passing the septa through the contacts.

Sharp Tips on No. 212 Retainer

Sharp tips on a No. 212 retainer should be sufficiently dulled to prevent damaging cementum.

Incorrect Technique for Cutting Septa

During removal of the rubber dam, an incorrect technique for cutting the septa may result in cut tissue or torn septa. Stretching the septa away from the gingiva, protecting the lip and cheek with an index finger and using curve-beaked scissors decreases the risk of cutting soft tissue or tearing the septa with the scissors as the septa are cut.

Cotton Roll Isolation and Cellulose Wafers

Absorbents such as cotton rolls (Fig. 8.32) also may provide isolation. Absorbents are isolation alternatives when rubber dam application is impractical or impossible. In selected situations, cotton roll isolation may be as effective as rubber dam isolation.[2,31] In conjunction with profound anaesthesia, absorbents provide acceptable moisture control for most clinical procedures. Using high-volume evacuation and/or a saliva ejector in conjunction with absorbents may abate salivary flow further. Cotton rolls should be replaced as needed. It is sometimes permissible to suction the free moisture from a saturated cotton roll while it is in place and thereby extend its use; this is done by placing the evacuator tip next to the end of the cotton roll while the operator secures the roll.

Several commercial devices for holding cotton rolls in position are available (Fig. 8.33). It is generally necessary to remove the holding appliance from the mouth to change the cotton rolls. An advantage of cotton roll holders is that they may slightly retract the cheeks and tongue from teeth, which enhances access and visibility.

Placing a cotton roll in the facial vestibule (Fig. 8.34) isolates maxillary teeth. Placing a cotton roll in the vestibule and another between teeth and the tongue (Fig. 8.35) isolates mandibular teeth. Although placement of a cotton roll in the facial vestibule is simple, placement on

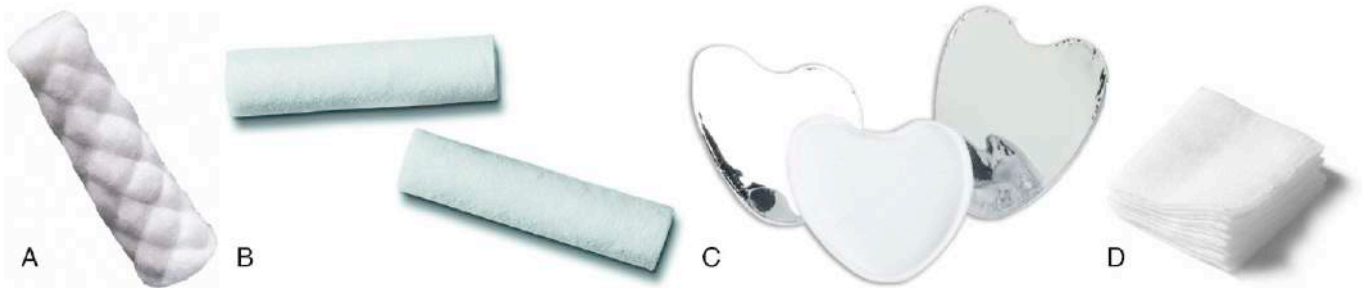


Figure 8.32 Absorbents such as cotton rolls (A and B), reflective shields (C) and gauze sponges (D) provide satisfactory dryness for short periods.
(Courtesy: Richmond Dental, Charlotte, NC.)



Figure 8.33 A cotton roll holder in position.
(Courtesy: R. Scott Eidson, DDS.)



Figure 8.34 Isolate maxillary posterior teeth by placing the cotton roll in the vestibule adjacent to teeth.
(Courtesy: R. Scott Eidson, DDS.)

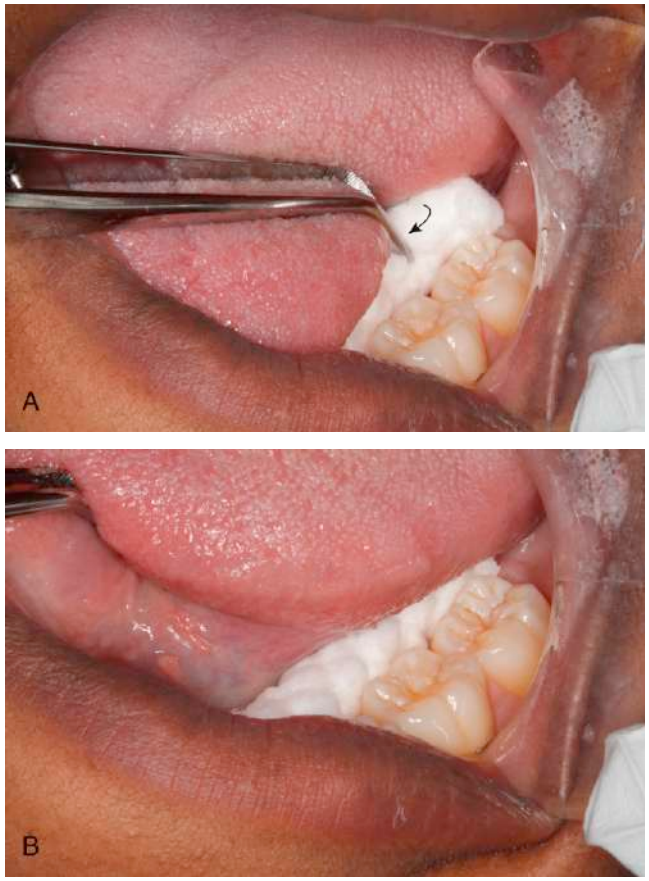


Figure 8.35 (A) Position a large cotton roll between the tongue and teeth by 'rolling' the cotton to place it in the direction of the arrow. (B) Properly positioned facial and lingual cotton rolls improve access and visibility.
(Courtesy: R. Scott Eidson, DDS.)

the lingual of mandibular teeth is more difficult. Lingual placement is facilitated by holding the mesial end of the cotton roll with operative pliers and positioning the cotton roll over the desired location. The index finger of the other hand is used to push the cotton roll gingivally while twisting the cotton roll with the operative pliers towards the lingual aspect of teeth. Cellulose wafers may be used to retract the cheek and provide additional absorbency. After the cotton rolls, cellulose wafers or both are in place, the saliva ejector may be positioned. When removing cotton rolls or cellulose wafers, it may be necessary to moisten them using the air–water syringe to prevent inadvertent removal of the epithelium from the cheeks, floor of the mouth or lips.

Other Isolation Techniques

Throat Shields

When the rubber dam is not being used, throat shields are indicated when the risk of aspirating or swallowing small objects is present. Throat shields are particularly important when treating teeth in the maxillary arch. A gauze sponge (5 × 5 cm), unfolded and spread over the tongue and the posterior part of the mouth, is helpful in recovering a small object (e.g. an indirect restoration) should it be dropped (Fig. 8.36). It is possible for a small object to be aspirated or swallowed if a throat shield is not used (Fig. 8.37).[32]

High-volume Evacuators and Saliva Ejectors

Air–water spray is supplied through the head of the high-speed handpiece to wash the operating site and act as a coolant for the bur and the tooth. High-volume evacuators are preferred for suctioning water and debris from the mouth (Fig. 8.38) because saliva ejectors remove water slowly and have little capacity for picking up solids. A practical test for the adequacy of a high-volume evacuator is to submerge the evacuator tip in a 150-ml cup of water. The water should disappear in approximately 1 s. The combined use of water spray or air–water spray and a high-volume evacuator during cutting procedures has the following advantages:

I. Cuttings of tooth and restorative material and other debris are removed from the operating site.
II. A clean operating field improves access and visibility.
III. Dehydration of oral tissues does not occur.
IV. Precious metals may be more readily salvaged if desired.



Figure 8.36 A throat screen is used during try-in and removal of indirect restorations.
(Courtesy: R. Scott Eidson, DDS.)

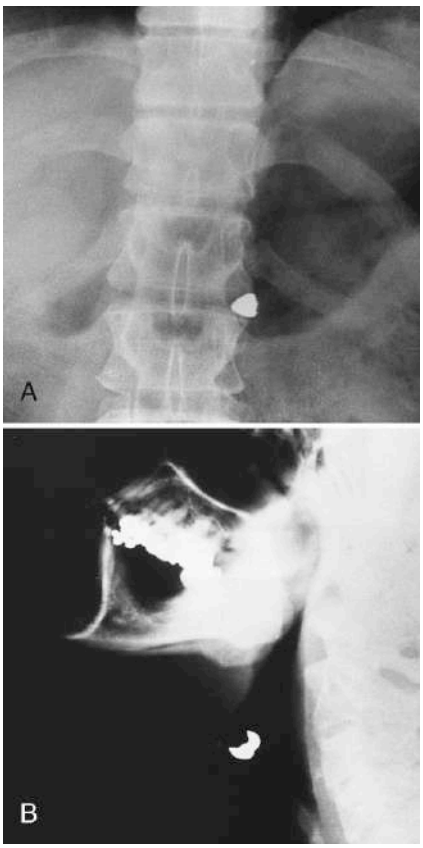


Figure 8.37 (A) Radiograph of swallowed casting in the patient's stomach. (B) Radiograph of casting lodged in the patient's throat.

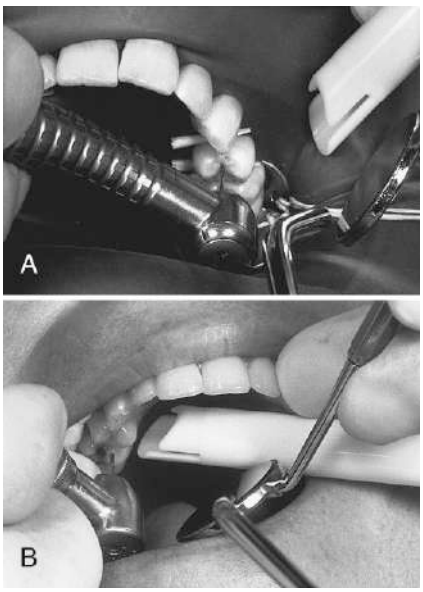


Figure 8.38 Position of evacuator tip for maximal removal of water and debris in operating area. (A) With rubber dam applied. (B) With cotton roll isolation.

The assistant places the evacuator tip as close as possible to the tooth being prepared. It should not, however, obstruct the operator's access or vision. Also, the evacuator tip should not be so close to the handpiece head that the air–water spray is diverted from the rotary instrument (i.e. bur or diamond). The assistant should place the evacuator tip in the mouth before the operator positions the handpiece and the mirror. The assistant usually places the tip of the evacuator just distal to the tooth to be prepared. For maximal efficiency, the orifice of the evacuator tip should be positioned such that it is parallel to the facial (lingual) surface of the tooth being prepared. The assistant's right hand holds the evacuator tip; the left hand manipulates the air–water syringe. (Hand positions are reversed if the operator is left-handed.) When the operator needs to examine the progress of tooth preparation, the assistant rinses and dries the tooth using air from the syringe in conjunction with the evacuator.

In most patients, the use of saliva ejectors is not required for removal of saliva because salivary flow is greatly reduced when the operating site is profoundly anaesthetized. The dentist or assistant positions the saliva ejector if needed. The saliva ejector removes saliva that collects on the floor of the mouth. It may be used in conjunction with sponges, cotton rolls and the rubber dam. It should be placed in an area least likely to interfere with the operator's movements.

The tip of the ejector must be nonabrasive. Disposable, adjustable plastic ejectors are preferable because of improved infection control (Fig. 8.39). The ejector should be placed to prevent occluding its tip with tissue from the floor of the mouth. Some ejectors are designed to prevent suctioning of tissue. It also may be necessary to adjust the suction for each patient to prevent this occurrence. *Svedopter* (saliva ejector with tongue retractor) moisture control systems, which aid in providing suction, retraction, illumination and jaw opening support, are available (Isolite Systems, Santa Barbara, CA, USA). A reduction in operating time when placing sealants has been reported when using the Isolite.[33] The same study reported that the majority of patients were indifferent with regard to isolation with Isolite or cotton rolls, considering both techniques comfortable.[33]

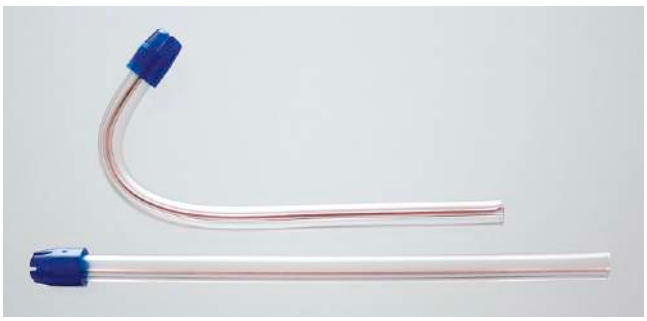


Figure 8.39 Saliva ejectors.
(From Boyd LRB: *Dental instruments: a pocket guide,* ed 4, St. Louis, MO, 2012, Saunders.)

Retraction Cord

When properly applied, retraction cord often may be used for isolation and retraction in direct procedures involving accessible subgingival areas and in indirect procedures involving gingival margins. When the rubber dam is not used, is impractical or is inappropriate, retraction cord, usually moistened with a noncaustic haemostatic agent, may be placed in the gingival sulcus to displace the gingiva and allow local control of sulcular seepage and haemorrhage. To achieve adequate moisture control, retraction cord isolation should be used in conjunction with salivation control. A properly applied retraction cord improves access and visibility and helps prevent abrasion of gingival tissue during tooth preparation. Retraction cord may help limit excess restorative material from entering the gingival sulcus and provide better access for contouring and finishing the restorative material. Anaesthesia of the operating site may or may not be needed for patient comfort.

Clinical technique:

- The operator chooses a *diameter of cord* that will fit in the gingival sulcus and cause lateral displacement of the free gingiva (opening the sulcus) without 'blanching' it (i.e. without causing tissue ischaemia secondary to pressure from the cord).
- The *length of the cord* should be sufficient to extend approximately 1 mm beyond the gingival width of the tooth preparation.
- A *thin, blunt-edged instrument blade* or the side of an explorer is used to progressively insert the cord. To prevent dislodgment of previously inserted cord, the placement instrument should be moved slightly backward at each step as it is stepped along the cord (Fig. 8.40).

Cord placement should not harm gingival tissue or damage the epithelial attachment. If ischaemia of gingival tissue is observed, the cord may need to be replaced with a smaller diameter cord. The objective is to obtain minimal yet sufficient lateral displacement of the free gingiva and not to force it apically. Cord insertion results in adequate displacement of the gingival crest in a short time. Occasionally it may be helpful to insert a second, usually larger, cord over the initially inserted cord.

In procedures for an indirect restoration, inserting the cord before removal of infected dentin and placement of any necessary liner assists in providing maximum moisture control. It also opens the sulcus in readiness for any bevelling of the gingival margins, when indicated. The cord may be removed before bevelling or it may be left in place during bevelling. Inserting the cord as early as possible in tooth preparation helps prevent abrasion of the gingival tissue, thus reducing the potential for bleeding and allowing only minimal absorption of any medicament from the cord into the circulatory system.



Figure 8.40 Retraction cord placed in the gingival crevice. (A) Cord placement initiated. (B) A thin, flat-bladed instrument is used for cord placement. (C) Cord placed.

Mirror and Evacuator Tip Retraction

A secondary function of the mirror and the evacuator tip is to retract the cheek, lip and tongue (Fig. 8.41). This retraction is particularly important when a rubber dam is not used.

Mouth Props

A potential aid to restorative procedures on posterior teeth (for a lengthy appointment) is a mouth prop (Fig. 8.42A and B). A prop should establish and maintain suitable mouth opening. Its use may also help relieve masticatory muscle fatigue. The ideal characteristics of a mouth prop are as follows:

I. Adaptable to all mouths.
II. Easily positioned, without causing discomfort to the patient.
III. Readily adjusted, if necessary, to provide the proper mouth opening or improve its position in the mouth.
IV. Stable once applied.
V. Rapidly removed in case of emergency.
VI. Sterilizable or disposable.

Mouth props are generally available as either a block type or a ratchet type (Fig. 8.42C–E). Although the ratchet type is adjustable, its size and cost are disadvantages.

The use of a mouth prop may be beneficial to the operator and the patient. The most outstanding benefits to the patient are relief of responsibility of maintaining adequate mouth opening and relief of muscle fatigue and associated discomfort. For the dentist, the prop ensures constant and adequate mouth opening and permits extended or multiple operations, if desired.



Figure 8.41 Chair-side assistant uses air syringe to retract the lip while teeth dry to keep the mirror clear.

Drugs

The use of drugs to control salivation is rarely indicated in restorative dentistry and is generally limited to *atropine*. As with any drug, the operator should be familiar with its indications, contraindications and adverse effects. Atropine is contraindicated for nursing mothers and patients with glaucoma.[34]

Summary

A thorough knowledge of the preliminary procedures addressed in this chapter affords maximum comfort for the patient while reducing physical strain on the dental team. Maintaining optimal moisture control is a necessary component in the delivery of high-quality operative dentistry.

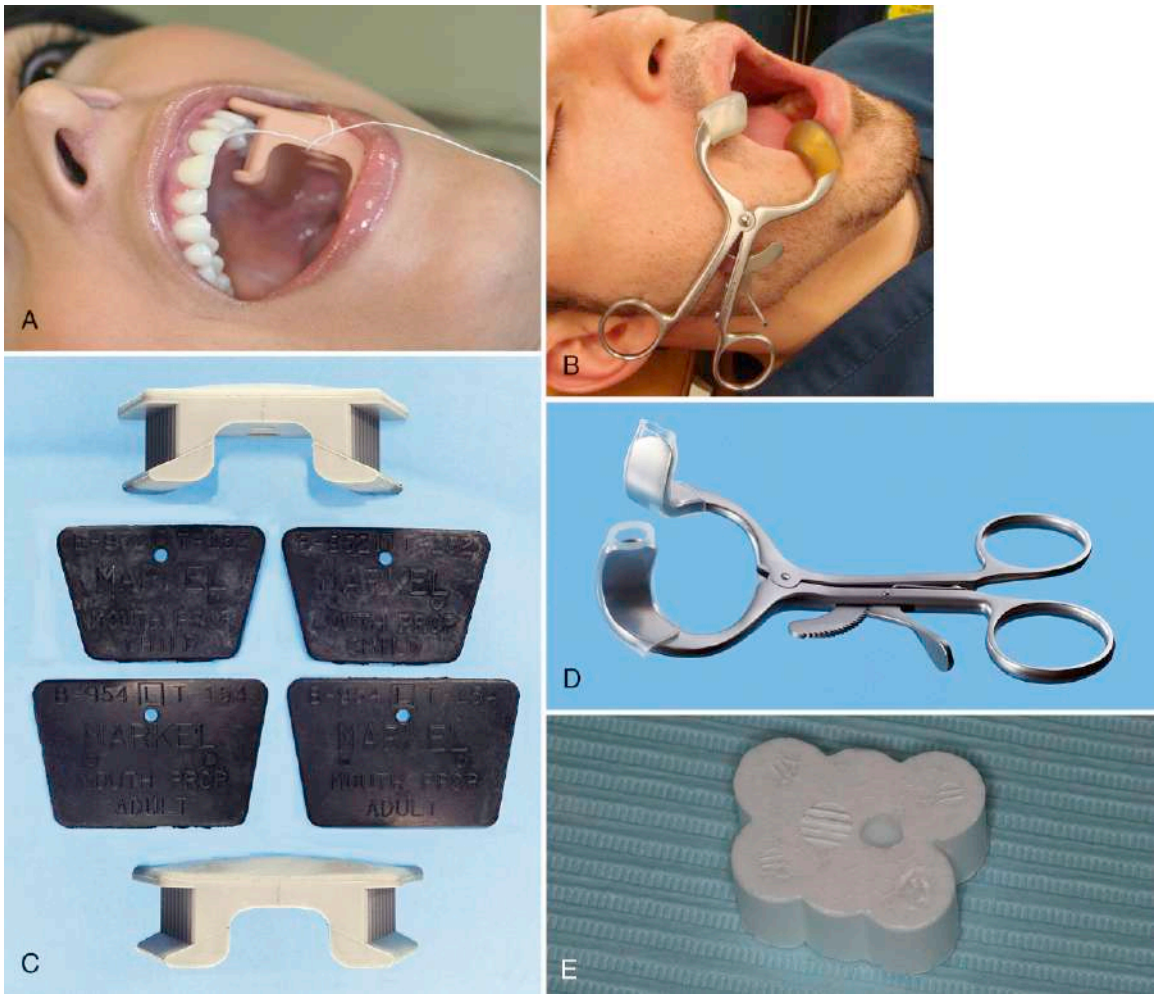


Figure 8.42 Mouth props. (A) Block-type prop maintaining mouth opening. (B) Ratchet-type prop maintaining mouth opening. (C) Block-type prop. (D) Ratchet-type prop. (E) Foam-type disposable prop.
([A and B] From Malamed SF: *Sedation: a guide to patient management,* ed 5, St. Louis, MO, 2010, Mosby. [C and D] From Hupp JR, Ellis E, Tucker MR: *Contemporary oral and maxillofacial surgery,* ed 5, St. Louis, MO, 2008, Mosby.)

References

1. Shugars DA, Williams D, Cline SJ, et al: Musculoskeletal back pain among dentists, *Gen Dent* 32:481–485, 1984.
2. Raskin A, Setcos JC, Vreven J, et al: Influence of the isolation method on the 10-year clinical behaviour of posterior resin composite restorations, *Clin Oral Invest* 25:148–152, 2000.
3. Fusayama T: Total etch technique and cavity isolation, *J Esthet Dent* 4:105–109, 1992.
4. Heling I, Sommer M, Kot I: Rubber dam—an essential safeguard, *Quintessence Int* 19:377–378, 1988.
5. Huggins DR: The rubber dam—an insurance policy against litigation, *J Indiana Dent Assoc* 65:23–24, 1986.
6. Anusavice KJ, editor: *Phillips' science of dental materials*, ed 11, Saunders, 2003, St. Louis, MO.
7. Medina JE: The rubber dam—an incentive for excellence, *Dent Clin North Am*:255–264, 1967.
8. Christensen GJ: Using rubber dams to boost quality, quantity of restorative services, *J Am Dent Assoc* 125:81–82, 1994.
9. American Dental Association Council on Scientific Affairs: ADA Council on Dental Benefit Programs: statement on posterior resin-based composites, *J Am Dent Assoc* 129:1627–1628, 1998.
10. Barghi N, Knight GT, Berry TG: Comparing two methods of moisture control in bonding to enamel: a clinical study, *Oper Dent* 16:130–135, 1991.
11. Smales RJ: Rubber dam usage related to restoration quality and survival, *Br Dent J* 174:330–333, 1993.
12. Roy A, Epstein J, Onno E: Latex allergies in dentistry: recognition and recommendations, *J Can Dent Assoc* 63:297–300, 1997.
13. Albani F, Ballesio I, Campanella V, et al: Pit and fissure sealants: results at five and ten years, *Eur J Paediatr Dent* 6:61–65, 2005.
14. Nimmo A, Werley MS, Martin JS, et al: Particulate inhalation during the removal of amalgam restorations, *J Prosthet Dent* 63:228–233, 1990.
15. Berglund A, Molin M: Mercury levels in plasma and urine after removal of all amalgam restorations: the effect of using rubber dams, *Dent Mater* 13:297–304, 1997.
16. Kremers L, Halbach S, Willruth H, et al: Effect of rubber dam on mercury exposure during amalgam removal, *Eur J Oral Sci* 107:202–207, 1999.
17. Cochran MA, Miller CH, Sheldrake MA: The efficacy of the rubber dam as a barrier to the spread of microorganisms during dental treatment, *J Am Dent Assoc* 119:141–144, 1989.
18. Samaranayake LP, Reid J, Evans D: The efficacy of rubber dam isolation in reducing atmospheric bacterial contamination, *ASDC J Dent Child* 56:442–444, 1989.
19. Harrel SK, Molinari J: Aerosols and splatter in dentistry: a brief review of the literature and infection control implications, *J Am Dent Assoc* 135:429–437, 2004.
20. Joynt RB, Davis EL, Schreier PH: Rubber dam usage among practicing dentists, *Oper Dent* 14:176–181, 1989.
21. Marshall K, Page J: The use of rubber dam in the UK: a survey, *Br Dent J* 169:286–291, 1990.
22. Gilbert GH, Litaker MS, Pihlstrom DJ, et al: DPBRN Collaborative Group: rubber dam use during routine operative dentistry procedures: findings from the dental PBRN, *Oper Dent* 35:491–499, 2010.
23. de Andrade ED, Ranali J, Volpato MC, et al: Allergic reaction after rubber dam placement, *J Endod* 26:182–183, 2000.
24. Jones CM, Reid JS: Patient and operator attitudes toward rubber dam, *ASDC J Dent Child* 55:452–454, 1988.
25. Peterson JE, Nation WA, Matsson L: Effect of a rubber dam clamp (retainer) on cementum and junctional epithelium, *Oper Dent* 11:42–45, 1986.
26. Ingraham R, Koser JR: *An atlas of gold foil and rubber dam procedures*, Buena Park, CA, 1961, Uni-Tro College Press.
27. Brinker HA: Access—the key to success, *J Prosthet Dent* 28:391–401, 1972.
28. Cunningham PR, Ferguson GW: The instruction of rubber dam technique, *J Am Acad Gold Foil Oper* 13:5–12, 1970.
29. Markley MR: Amalgam restorations for Class V cavities, *J Am Dent Assoc* 50:301–309, 1955.
30. Baum L, Phillips RW, Lund MR: *Textbook of operative dentistry*, ed 3, Philadelphia, PA, 1995, Saunders.
31. Brunthaler A, König F, Lucas T, et al: Longevity of direct resin composite restorations in posterior teeth, *Clin Oral Invest* 7:63–70, 2003.
32. Nelson JF: Ingesting an onlay: a case report, *J Am Dent Assoc* 123:73–74, 1992.
33. Collette J, Wilson S, Sullivan D: A study of the Isolite system during sealant placement: efficacy and patient acceptance, *Pediatr Dent* 32:146–150, 2010.
34. Ciancio SG, editor: *ADA/PDR dental therapeutics*, ed 5, PDR Network, 2009.